

MOTION IN A STRAIGHT LINE

Motion: If a body changes its position with time body is called in motion

★ Rest and motion are relative terms. They depend on observer.

Jaise ki yahan babu bhaiya aapke respect mein motion mein hain, but thele ke respect mein rest par hain.



Distance: Actual path travelled by the body -

- ★ Dependent upon path (we should know the path)
- ★ It can't be decreasing
- ★ It can't be negative
- ★ It can be zero or positive

DISPLACEMENT

Change in position vector $\vec{r}$ or change in position

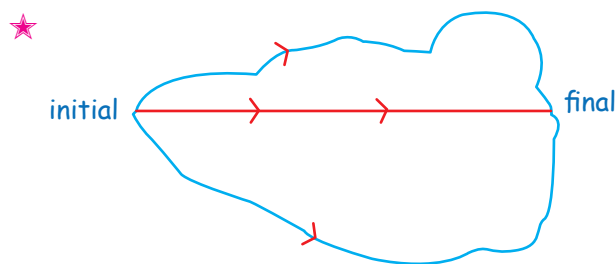
$$\vec{d} = \Delta \vec{r} = \vec{r}_f - \vec{r}_i$$

- ★ Displacement = final position - Initial position
- ★ It is the shortest distance between initial and final point.
- ★ Vector term
- ★ It can be positive, negative or zero.
- ★ It can be increasing or decreasing.
- ★ Independent of path travelled.

Q. A particle move from point A to point B. Find its displacement in following case.

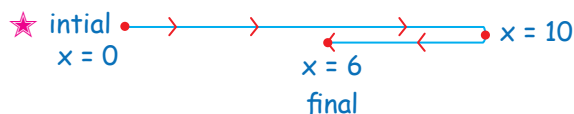


Sol. $\vec{d} = \text{displacement} = 5\hat{i} + 2\hat{j} + 5\hat{k}$



Displacement - same

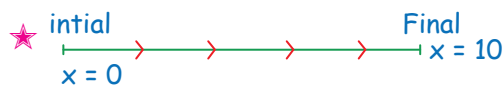
Distance - different (depend upon path)



distance $\rightarrow 10 + 4 = 14$

displacement $\rightarrow 6$

(vector form) $\rightarrow 6\hat{i}$



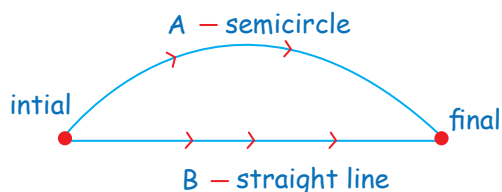
Distance = 10

Displacement = $10\hat{i}$

Here, Distance = |Displacement|

- ★ |Displacement| = magnitude of displacement
- ★ If a particle does not change direction then distance = |displacement|
- ★ Distance $\geq$ |displacement|
- ★ $\frac{\text{Distance}}{|\text{Displacement}|} \geq 1$

Q. A particle move from initial to final position in two different path A and B as shown in fig. Find distance and displacement for both.



Sol.

	A	B
Distance	πR	$2R$
Displacement	$2R$	$2R$

Q. A particle move 5m along east then 6m along north and 10m in upward direction. Find distance & displacement?

Sol. Distance $5\text{m} + 6\text{m} + 10\text{m} = 21$

$$\text{Displacement} = 5\hat{i} + 6\hat{j} + 10\hat{k}$$

$$\text{Magnitude} = \sqrt{5^2 + 6^2 + 10^2}$$

Q. A particle move

$$10\text{m east} = 10\hat{i}$$

$$5\text{m north} = 5\hat{j}$$

$$6\text{m south} = -6\hat{j}$$

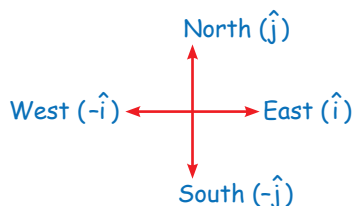
$$8\text{m west} = -8\hat{i}$$

$$15\text{m east} = 15\hat{i}$$

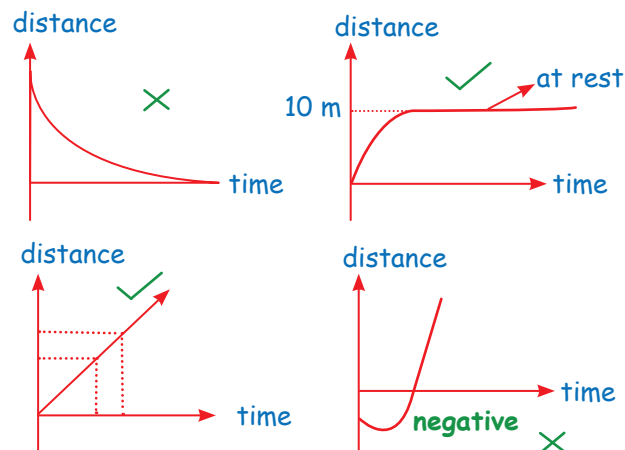
$$20\text{m north} = 20\hat{j}$$

$$\text{distance} = 64\text{ m}$$

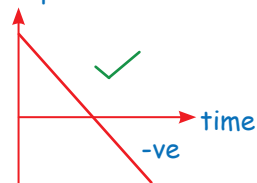
$$\begin{aligned} \text{displacement} &= (10 - 8 + 15)\hat{i} + (5 - 6 + 20)\hat{j} \\ &= 17\hat{i} + 19\hat{j} \end{aligned}$$



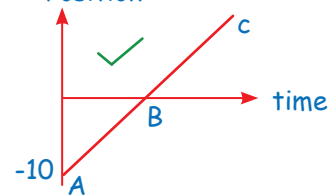
Q. Which of the graph is possible.



Displacement



Position



displacement = change in position.

Average velocity = $\frac{\text{total displacement}}{\text{total time}}$

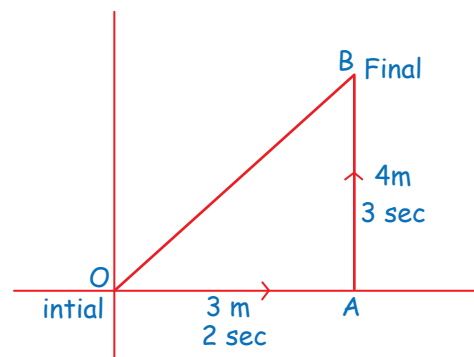
Instantaneous velocity = $\frac{dr}{dt}$

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Instantaneous speed = kisi inst. par speed

|Instantaneous velocity| = Instantaneous speed

Q. A particle move from origin to point A and took 2 sec, then it move to point B and took 3 sec in following fig. Find displacement, average velocity and average speed.



Sol. Displacement $\vec{d} = 3\hat{i} + 4\hat{j}$

$$\text{Avg. velocity} = \frac{3\hat{i} + 4\hat{j}}{5}$$

$$\text{Avg. speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{3 + 4}{2 + 3} = \frac{7}{5}$$

Q. $x = 0$ $\xrightarrow{\quad\quad\quad}$ $x = 10$
 $t = 10\text{ sec}$
 $x = 6$

Sol. Displacement = 6 (mag)

$$\text{Average velocity} = \frac{6}{10}$$

$$\text{Distance} = 10 + 4 = 14$$

$$\text{Average speed} = \frac{14}{10}$$

Q. $x = 0$ 

Sol. Distance = 10

$$\text{Average speed} = \frac{10}{5}$$

$$\text{Displacement} = 10 \text{ (Magnitude)}$$

$$\text{Average velocity} = \frac{10}{5} \text{ (Magnitude)}$$

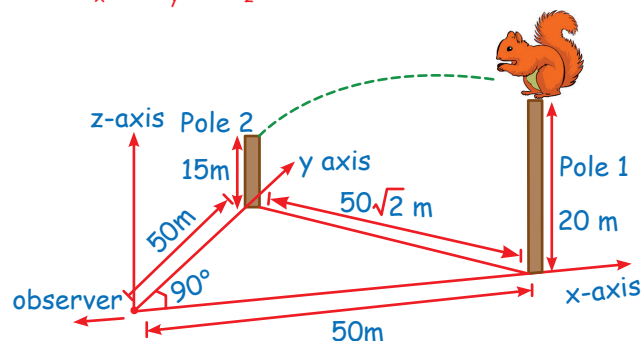
★ Agar particle ne apni direction nahi badli

★ distance = |displacement|

★ Avrg speed = |Avrg velocity|



Q. A small squirrel jumps from pole 1 to pole 2 and took 3 sec. What is average velocity vector of squirrel? If average velocity vector is expressed as $v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$, express your answer as sum of magnitudes of its components $|v_x| + |v_y| + |v_z|$ in unit m/s.



Sol. Initial coordinate is (50, 0, 20)

Final coordinate is (0, 50, 15)

$$\text{Displacement} = -50\hat{i} + 50\hat{j} - 5\hat{k}$$

$$\text{Average velocity} = \frac{-50\hat{i} + 50\hat{j} - 5\hat{k}}{3}$$

$$= -\frac{50}{3}\hat{i} + \frac{50}{3}\hat{j} - \frac{5}{3}\hat{k}$$

$$\frac{50}{3} + \frac{50}{3} + \frac{5}{3} = \frac{105}{3} = 35 \text{ m/s}$$

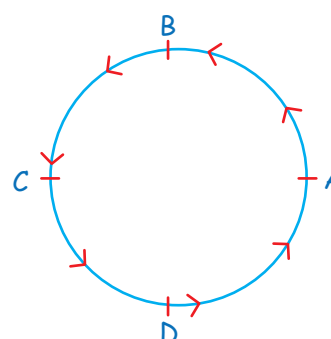
Mujhe pata hai kuch google boys ne google par search karke answer 105 nikala hoga... Abe, sawal theek se padh liya karo.



Padh toh be kaa likha hai



Q. A particle is performing uniform circular motion with constant speed v_1 having time period T Anticlockwise. Find avg velocity and avg speed.



Sol.

	Avg Speed	Avg Velocity
$A \rightarrow B$	$\frac{2\pi R / 4}{T / 4} = \frac{2\pi R}{T}$	$\frac{R\sqrt{2}}{T / 4}$
$A \rightarrow B \rightarrow C$	$\frac{\pi R}{T / 2} = \frac{2\pi R}{T}$	$\frac{2R}{T / 2}$
$A \rightarrow B \rightarrow C \rightarrow D$	$\frac{(3/4)2\pi R}{3T / 4} = \frac{2\pi R}{T}$	$\frac{R\sqrt{2}}{3T / 4}$
$A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$	$\frac{2\pi R}{T}$	0

Q. A car is moving along x-axis, in 1st four hour it travel with speed 50 km/hr, in next 2 hours it move with 70 km/hr and in last part of journey it travel for 5 hour with 80 km/hr. Find avg speed.

Sol. Avg speed = $\frac{\text{total distance}}{\text{Time}}$

$$= \frac{d_1 + d_2 + d_3}{t_1 + t_2 + t_3} = \frac{50 \times 4 + 70 \times 2 + 80 \times 5}{4 + 2 + 5}$$

$$= \frac{200 + 140 + 400}{11}$$

$$\text{Averg speed} = \frac{740}{11}$$

note: 1D motion (on x-axis)

$$\text{Average velocity} = \frac{\vec{x}_f - \vec{x}_i}{\text{time}} = \frac{\Delta \vec{x}}{\Delta t}$$

Main average value ko represent karne ke kuch symbol neeche likh raha hoon jo aksar use hote hain aur space bachane ke liye hum bhi use karenge.

$$\text{Average velocity} = \langle \vec{v} \rangle$$

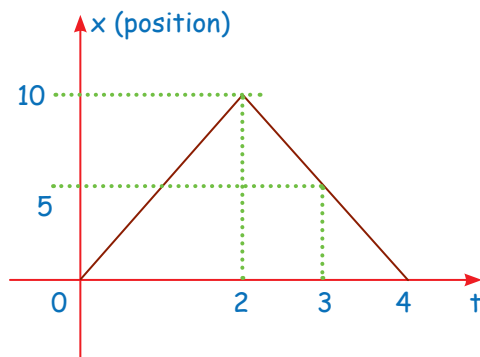
$$\text{Average speed} = \langle \text{speed} \rangle = \langle |\vec{v}| \rangle$$

$$\text{Average acc} = \langle \vec{a} \rangle$$

$$\text{Average kaddu} = \langle \text{kaddu} \rangle$$



Q. A particle is moving on the x-axis as show its x-coordinate unit time. Find average velocity from $t = 0$ to $t = 2$ sec, from $t = 0$ to $t = 3$ sec.



Sol. ① $t = 0 \rightarrow t = 2$ sec

$$\text{Average velocity} = \frac{x_f - x_i}{\text{time}} = \frac{10 - 0}{2} = 5$$

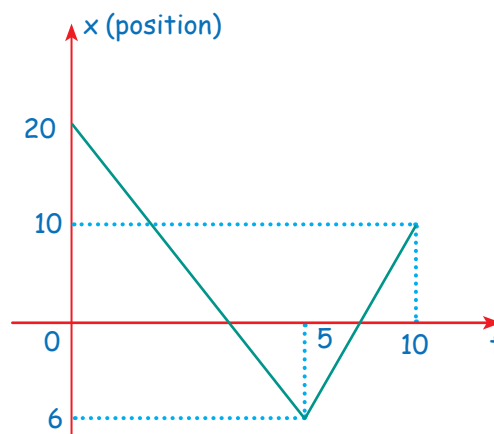
$$\text{Average speed} = \frac{10}{2} = 5 \left| \frac{10_f - 0_i}{t} \right|$$

② $t = 0 \rightarrow t = 3$

$$\text{Average velocity} = \frac{x_f - x_i}{\text{time}} = \frac{5 - 0}{3}$$

$$\text{Average speed} = \frac{10 + 5}{3}$$

Q. Find average velocity from $t = 0$ to $t = 5$ s and $t = 0$ to $t = 10$ s.



Sol. 1. $t = 0 \rightarrow t = 5$

$$\text{Average velocity} = \frac{x_f - x_i}{\text{time}} = \frac{(-6) - (+20)}{5}$$

$$\text{Average speed} = \frac{20 + 6}{5}$$

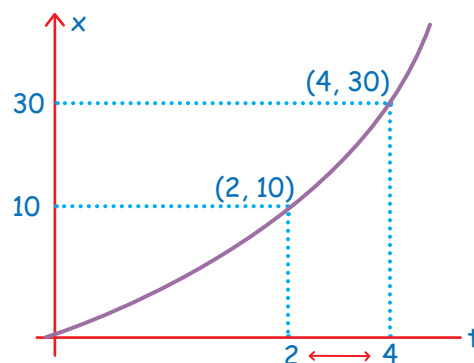
2. From $t = 0 \rightarrow t = 10$

$$\text{Average velocity} = \frac{x_f - x_i}{\text{time}} = \frac{10 - 20}{10} = \frac{-10}{10}$$

$$\text{Average speed} = \frac{20 + 6 + 6 + 10}{10} = 4.2 \text{ m/s}$$

Q. Find average velocity from

$t = 2 \rightarrow t = 4$ sec



Sol. Average velocity (mag) = $\frac{x_f - x_i}{\Delta t}$

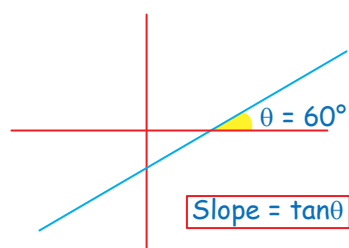
$$= \frac{20}{2} = 10$$

$$\text{Slope of line} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{30 - 10}{4 - 2}$$

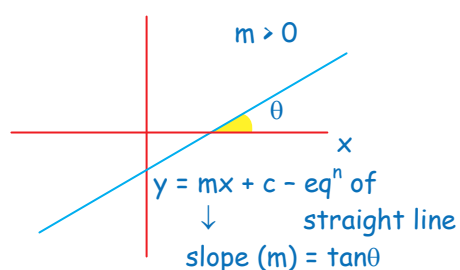
$$= \frac{20}{2} = 10$$

Chalo ab thoda sa kaam ka mathematics ka revision ho jaye jiski humein yahan bahut zaroorat hai.

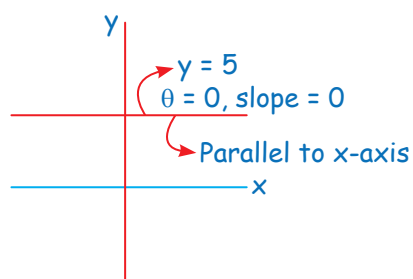
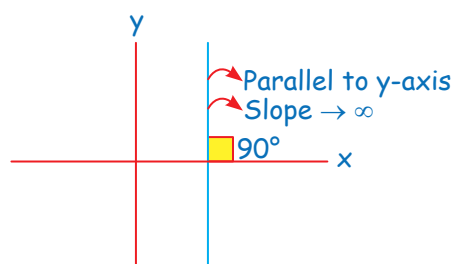
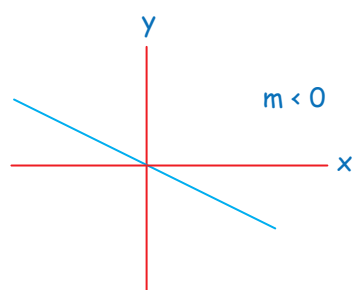
SLOPE OF LINE



$$\text{slope} = \sqrt{3} = \tan 60^\circ$$



$$\text{slope} = \tan \theta$$

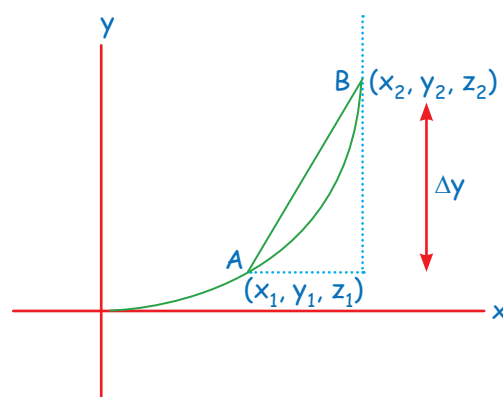


Slope of line join A to B

$$= \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \tan \theta$$

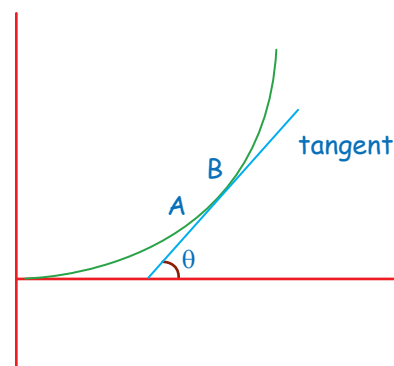
Δy = change in y

Δx = change in x



If Δy is very small $\Delta y \rightarrow 0$ ($\Delta y = dy$)

If Δx is very small $\Delta x \rightarrow 0$ ($\Delta x = dx$)



$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

★ $\frac{dy}{dx}$ matlab differentiation of y wrt x

→ aur rate of change of y wrt x

→ Slope of tangent at that point

→ Ratio of very-2 small change in y to very - 2 small change in x

→ 1st derivation of y wrt x

IMP FORMULA

$\frac{d}{dx}(x^n) = nx^{n-1}$	$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
$\frac{d}{dx} \sin x = \cos x$	$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
$\frac{d}{dx}(\cos x) = -\sin x$	$\frac{d}{dx} e^x = e^x$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\frac{d}{dx}(\ln x) = \frac{1}{x}$
$\frac{d}{dx}(\sec x) = \sec x \tan x$	

Q. $y = x^3$

Sol. $\frac{dy}{dx} = 3x^{3-1} = 3x^2$

Q. $y = t^3$

Sol. $\frac{dy}{dt} = 3t^2$

Q. $y = 2x^5$

Sol. $\frac{dy}{dx} = 2 \times 5x^4 = 10x^4$

Q. $y = \frac{1}{x^2} = x^{-2}$

Sol. $\frac{dy}{dx} = -2x^{-2-1} = \frac{-2}{x^3}$

Q. $y = x^2 + x^5$

Sol. $\frac{dy}{dx} = 2x + 5x^4$

Q. $y = x^9 + \frac{1}{x^5}$

Sol. $\frac{dy}{dx} = 9x^8 - \frac{5}{x^6}$

$\frac{dy}{dx} = 9x^8 - \frac{5}{x^6}$

differentiation of y wrt x

Q. $y = x$

Sol. $\frac{dy}{dx} = 1$

Q. $y = 5$ constant

Sol. $\frac{dy}{dx} = 0$

Q. $y = x^3 + \sin x$

Sol. $\frac{dy}{dx} = 3x^2 + \cos x$

Q. $y = x^7 + \tan x + 10$

Sol. $\frac{dy}{dx} = 7x^6 + \sec^2 x + 0$

Q. $y = 3x^3 + \sin x + \tan x$

Sol. $\frac{dy}{dx} = 9x^2 + \cos x + \sec^2 x$

Q. $y = 2x^2 + \cos x + 5$

Sol. $\frac{dy}{dx} = 4x - \sin x + 0$

Q. $y = 3x^2 + \cos x + e^x - \sin x + 10$

Sol. $\frac{dy}{dx} = 6x - \sin x + e^x - \cos x + 0$

Q. If $y = x^2 + 4x^{-1/2} - 3x^{-2}$ find $\frac{dy}{dx}$

Sol. $\frac{dy}{dx} = \frac{d}{dx}(x^2 + 4x^{-1/2} - 3x^{-2})$
 $= \frac{d}{dx}(x^2) + \frac{d}{dx}(4x^{-1/2}) - \frac{d}{dx}(3x^{-2})$
 $= \frac{d}{dx}(x^2) + 4 \frac{d}{dx}(x^{-1/2}) - 3 \frac{d}{dx}(x^{-2})$
 $= 2x - 2x^{-3/2} + 6x^{-3}$

Q. If $3y = 4x^2 - 5$ find $\frac{dy}{dx}$

Sol. $y = \frac{4}{3}x^2 - \frac{5}{3} \Rightarrow \frac{dy}{dx} = \frac{8x}{3}$

Q. $\sqrt{y} = x - 1$, find $\frac{dy}{dx}$

Sol. $y = x^2 - 2x + 1 \Rightarrow \frac{dy}{dx} = 2x - 2$

$\frac{dy}{dx}$ ko shortcut me y' bhi likhte hain. iska matlab ye hai ki y ka ek bar differentiation, usi prakar agar y ka do bar differentiation karenge to uss se

y'' ya $\frac{d^2y}{dx^2}$ likhte hai jaise

$y = x^3$
 $y' = \frac{dy}{dx} = 3x^2$
 $y'' = \frac{d^2y}{dx^2} = 6x$



Multiply

$y = u \cdot v$

$y' = uv' + vu'$

Q. $y = x^3 \cdot \sin x$

Sol. $\frac{dy}{dx} = x^3 \left(\frac{d}{dx} \sin x \right) + \sin x \left(\frac{d}{dx} x^3 \right)$

Q. $y = x^5 \tan x$

Sol. $\frac{dy}{dx} = x^5 \cdot \sec^2 x + [\tan x](5x^4)$

Q. $y = x^3 \cdot e^x$

Sol. $\frac{dy}{dx} = x^3 e^x + e^x \cdot 3x^2$

Q. $y = e^x \cdot \sin x$

Sol. $\frac{dx}{dy} = e^x \cdot \cos x + \sin x \cdot e^x$

Q. $y = x^3 e^x$

Sol. $\frac{dy}{dx} = x^3 e^x + e^x \cdot 3x^2$

Q. $y = x^4 \ln x$

Sol. $\frac{dy}{dx} = x^4 \cdot \frac{1}{x} + (\ln x)(4x^3)$

CHAIN RULE

Q. $y = \sin(x^2 + x^7)$

Sol. $\frac{dx}{dy} = \cos(x^2 + x^7) \times (2x + 7x^6)$

Upar wale chain rule ke question mein ye pattern pehchanne ki koshish karo ki pehle bahar wale function ka differentiation karna hai, fir multiply karte hue andar wale function ka differentiation karte jaana hai. Jaise

$y = \sin(\text{kaddu})$

$\frac{dy}{dx} = \cos(\text{kaddu}) \times \frac{d}{dx}(\text{kaddu})$



Q. $y = \ln x^5$

Sol. $\frac{dy}{dx} = \frac{1}{x^5} \times 5x^4$

Q. $y = \ln x^4$

Sol. $\frac{dy}{dx} = \frac{1}{x^4} \times 4x^3$

Q. $y = \sin^3 x = (\sin x)^3$

Sol. $\frac{dy}{dx} = 3(\sin x)^2 \cdot \cos x$

Q. $y = \ln(\sin x)$

Sol. $\frac{dy}{dx} = \frac{1}{\sin x} \times \cos x$

Q. $y = \tan^2 x$

Sol. $y = (2 \tan x) \cdot \sec^2 x$

Q. $y = \sin^4 x = (\sin x)^4$

Sol. $\frac{dy}{dx} = 4(\sin x)^3 \cdot \cos x$

Q. $y = \sin(x^2 + x^3)$

Sol. $\frac{dy}{dx} = \cos(x^2 + x^3)(2x + 3x^2)$

Q. $y = \ln(\sin x^3)$

Sol. $\frac{dy}{dx} = \frac{1}{\sin x^3} \times \cos(x^3) \times [3x^2]$

Q. $y = \sin(e^x)$

Sol. $\frac{dy}{dx} = \cos(e^x) \cdot e^x$

★ $\frac{d}{dx} \sin(ax+b) = a \cos(ax+b)$

★ $\frac{d}{dx} \sin(2x+3) = \cos(2x+3) [2+0]$

★ $\frac{d}{dx} e^{2x+5} = e^{2x+5} (2+0)$

Q. $y = x^3$ find $\frac{dy}{dx}$ at $x = 2$

Sol. $y = x^3$

$\frac{dy}{dx} = 3x^2$

at $x = 2$, $\frac{dy}{dx} = 12$

Q. $x = t^2 - 4t + 20$ find value of $\frac{dx}{dt}, \frac{d^2x}{dt^2}$ at $t = 2$ sec

Sol. $\frac{dx}{dt} = 2t - 4 + 0 \Rightarrow$ at $t = 2$ sec, $\frac{dx}{dt} = 0$

$\frac{d^2x}{dt^2} = 2 \Rightarrow$ at $t = 2$ sec, $\frac{d^2x}{dt^2} = 2$

Chain Rule Power wale questions

Q. $y = \sin^2 x$

$\frac{dy}{dx} = 2 \sin x \cdot \frac{d}{dx}(\sin x) = 2 \sin x \cos x$

Q. $y = \sin^2(3x+4)$

$\frac{dy}{dx} = 2 \sin(3x+4) \cos(3x+4) \times 3$

Q. $y = [\ln(3x^2)]^3$

$$\frac{dy}{dx} = 3[\ln(3x^2)]^2 \times \frac{1}{3x^2} \times 6x$$

Q. $y = (x^2 + 4)^{\frac{1}{2}}$

$$\frac{dy}{dx} = \frac{1}{2}(x^2 + 4)^{-\frac{1}{2}} \times 2x$$

Very Important

1. $y = \pi x^3$

$$\frac{dy}{dx} = \pi \times 3x^2$$

2. $y = x^5$

$$\frac{dy}{dt} = 5x^4 \frac{dx}{dt}$$

3. $y = \sin x$

$$\frac{dy}{dt} = \cos x \frac{dx}{dt}$$

4. $A = \pi r^2$

$$\frac{dA}{dt} = \pi 2r \frac{dr}{dt}$$

5. $y = \frac{4}{3} \pi x^3$

$$\frac{dy}{dx} = \frac{4}{3} \pi 3x^2$$

Q. Radius of a circle changes wrt time with the rate of + 5 m/sec, find rate of change of area wrt time when radius is 10 m

Sol. $A = \pi r^2, \frac{dr}{dt} = 5$

$$\frac{dA}{dt} = \pi 2r \frac{dr}{dt} \Rightarrow \frac{dA}{dt} = \pi \times 2 \times 10 \times 5$$

$$\frac{dy}{dx} \rightarrow \text{Rate of change of } y \text{ wrt } x$$

$$\frac{dA}{dt} \rightarrow \text{Rate of change of Area with time}$$

$$\frac{dr}{dt} \rightarrow \text{Rate of change of radius wrt time}$$

Q. If radius of a sphere is increasing at the rate of 10 m/sec at what rate volume of sphere will change, when radius is 3 m

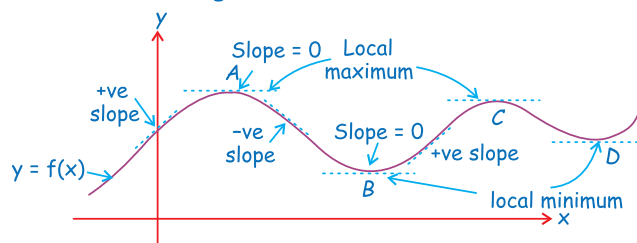
Sol. $V = \frac{4}{3} \pi r^3 - \frac{dv}{dt} = \frac{4\pi}{3} 3r^2 \frac{dr}{dt}$

$$\frac{dv}{dt} = \frac{4}{3} \pi 3(3)^2 \times 10 = 360\pi$$

MAXIMA AND MINIMA

Finding maxima and minima of a function using derivatives:-

A maximum is a high point and minimum is low point of a function (see figure)



In a smoothly changing function a maximum or a minimum is always where function flattens out or where slope of tangent line is zero. We know slope = $\frac{dy}{dx}$. So a function reaches its maximum or minimum value when $\frac{dy}{dx} = 0$.

In the neighbourhood of maximum (point A), slope changes from positive to zero at point A and then becomes negative as x increases which means

$$\frac{d}{dx}(\text{slope}) < 0 \Rightarrow \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} < 0$$

In the neighbourhood of minimum (point B), slope changes from negative to zero and then becomes positive as x increases which means

$$\frac{d}{dx}(\text{slope}) > 0 \Rightarrow \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} > 0$$

SECOND DERIVATIVE TEST

When a function's slope $\left(\frac{dy}{dx}\right) = 0$ at a point and its

second derivative at that point is

- (i) less than zero, it is a local maximum.
- (ii) greater than zero, it is a local minimum.

★ Jab bhi kabhi kisi function ki max ya min value chahiye hogi, hum use differentiate karke zero kar denge.

★ For point of maxima $\frac{dy}{dx} = 0, \frac{d^2y}{dx^2} < 0$

★ For point of minima $\frac{dy}{dx} = 0, \frac{d^2y}{dx^2} > 0$

Q. $y = x^2 - 4x + 20$

Find $y_{\min} = ?$

Sol. $\frac{dy}{dx} = 2x - 4 = 0 \Rightarrow x = 2$

$$\frac{d^2y}{dx^2} = 2 - 0 = 2 > 0$$

$$\frac{d^2y}{dx^2} > 0 \text{ hence at } x = 2, y \text{ is min}$$

$$\text{So, } y_{\min} = 2^2 - 4 \times 2 + 20 = 16$$

Q. $y = \sqrt{3} \sin \theta + \cos \theta$ Find $y_{\max}$

Sol. $\frac{dy}{d\theta} = \sqrt{3} \cos \theta - \sin \theta = 0$

$$\sqrt{3} \cos \theta = \sin \theta$$

$$\theta^\circ = 60^\circ$$

$$\frac{d^2y}{d\theta^2} = -\sqrt{3} \sin \theta - \cos \theta$$

$$\text{at } \theta = 60^\circ$$

$$\frac{d^2y}{d\theta^2} = \frac{-3}{2} - \frac{1}{2} < 0$$

So y is max.

$$y_{\max} = \sqrt{3} \sin 60^\circ + \cos 60^\circ$$

$$\sqrt{3} \left(\frac{\sqrt{3}}{2} \right) + \frac{1}{2} = 2$$

$$y = a \sin \theta + b \cos \theta$$

$$y_{\max} = \sqrt{a^2 + b^2}$$

$$y_{\max} = \sqrt{3+1} = 2$$

Q. What is the minimum value of y for the curve $y = -8x^2 + x^4$.

Sol. $y = -8x^2 + x^4$

$$\frac{dy}{dx} = -16x + 4x^3 = -x(16 - 4x^2)$$

The function will have a maximum or minimum value when $\frac{dy}{dx} = 0$

$$\Rightarrow x(16 - 4x^2) = 0$$

$$\Rightarrow x = 0 \text{ or } x = \pm 2$$

Now

$$\frac{d^2y}{dx^2} = -16 + 12x^2$$

$$\text{At } x = 0, \frac{d^2y}{dx^2} = -16 \text{ (maximum)}$$

$$\text{At } x = \pm 2, \frac{d^2y}{dx^2} = -16 + 48 = +32 \text{ (minimum)}$$

So function has minimum value at $x = \pm 2$

$$y_{\min} = -8 \times 4 + 16 = -16$$

Q. A ball is thrown vertically upward in the air. Its height y at any time t is given by $y = 10t - 5t^2$ where y is in meters and t is in seconds. What is the maximum height attained by the ball?

Sol. $y = 10t - 5t^2$

$$\frac{dy}{dt} = 10 - 10t = 0$$

$$\Rightarrow t = 1 \text{ sec}$$

$$\frac{d^2y}{dt^2} = -10 \text{ (maximum)}$$

ball attains maximum height at $t = 1s$

$$y_{\max} = 10 \times 1 - 5 \times 1^2 = 5m.$$

★ Inst. velocity = $\vec{v} = \frac{d\vec{r}}{dt}$

★ If particle moving on x-axis

$$\text{Inst. velocity} = \vec{v}_x = \frac{dx}{dt} \hat{i}$$

Seedhi baat x ko time ke respect me differentiate karenge to velocity aayegi.



Q. A particle is moving on x-axis such that its x-coordinate w.r.t time change as $x = t^2 - 6t + 10$

(a) Find velocity at $t = 0, t = 3, t = 6$ sec, = Inst velocity

Sol. $v = \frac{dx}{dt} = 2t - 6$
(now put the value of t)

$$t = 0, v = -6$$

$$t = 3, v = 0$$

$$t = 6, v = +6$$

★ Kisi time par v matlab instantaneous velocity.

★ Kisi time par velocity ka magnitude hi us time par instantaneous speed hai.

(b) Find avrg velocity between $t = 0 \rightarrow t = 3$

Sol. Average velocity = $\frac{x_f - x_i}{t_f - t_i}$

$$t = 0, x_i = 0 - 0 + 10 = 10$$

$$t = 3, x_f = 3^2 - 6 \times 3 + 10 = 1$$

$$\text{Average velocity} = \frac{1-10}{3} = -3\hat{i}$$

Definition carefully yaad rakhni hai, avg velocity ke liye x ko differentiate mat kar dena.

Avg velocity nikalne k liye bas ye dekho x_i aur x_f kya hai



Q. $x = t^2 - 2t + 10$. Find avg velocity and avg speed from $t = 0$ to $t = 3$ sec

Sol. At $t = 0, x = 10$

At $t = 3, x = 9 - 6 + 10 = 13$

$$\text{Avg velocity} = \frac{13-10}{3} = 1$$

Avg speed = 1 (Kyon bhai, tune yahi kiya na?)



Abe, galat hai! Avg speed (average speed) hamesha avg velocity ke barabar nahi hoti. Definition yaad karo, kya thi?



SKC

Avg speed ke liye humein rasta (path) pata hona zaroori hai. So, sabse pehle initial aur final position nikalo aur velocity zero karke turning point nikalo. Fir rasta (path) bana kar dekho particle kitna chala hai.



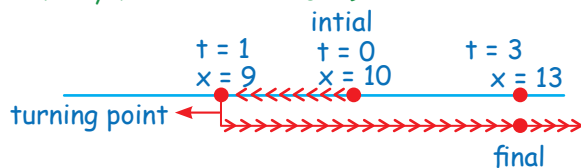
At $t = 0, x = 10$

At $t = 3, x = 13$ now find turning point

$$v = 2t - 2 = 0$$

$t = 1$ (dekho, is time par particle ne apni direction change ki hai, pata lagao $t = 1$ par particle kahan hai).

$$t = 1, x = 1^2 - 2 \times 1 + 10 = 9$$



$$\text{Average speed} = \frac{1+4}{3} = \frac{5}{3}$$

Better understanding k liye niche wale ques aur solve kro

Q. $x = 2t^2 - 4t + 5$ from $t = 0 \rightarrow t = 2$ Find Average speed

Sol. $t = 0, x_i = 5$

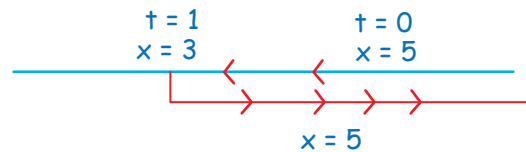
$t = 2, x_f = 5$

$$v = 4t - 4 = 0 \text{ (for turning point)}$$

$$4t = 4$$

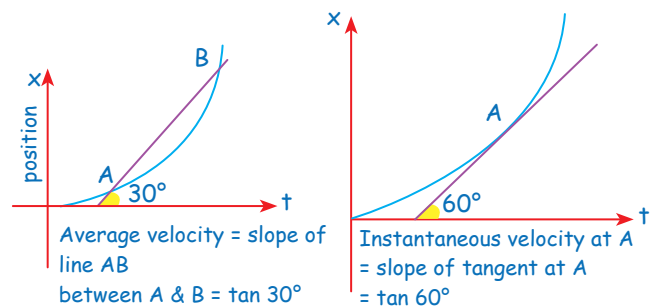
$$t = 1$$

At $t = 1, x = 2 - 4 + 5 = 3$

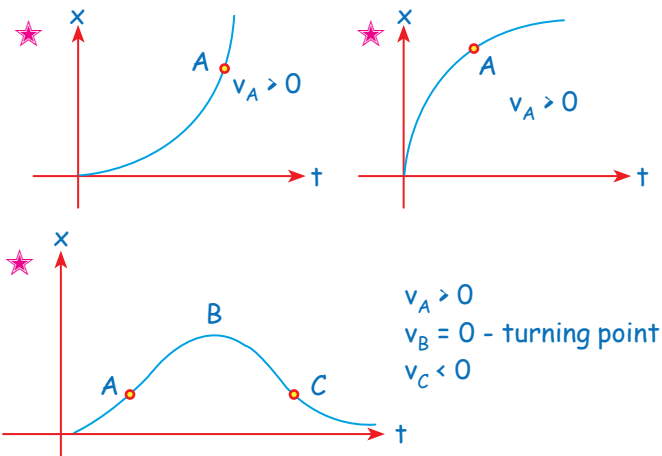


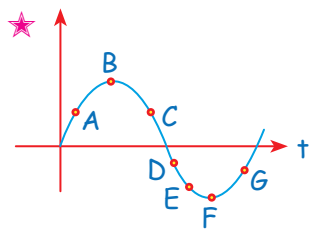
$$\text{Average speed} = \frac{2+2}{2} = 2 \text{ m/s}$$

GRAPHS



$x-t$ graph mein kisi point par tangent ka slope us point par velocity dega. Ise mat bhoalna.





$$\begin{aligned} v_A &> 0 \\ v_B &= 0 \\ v_C &< 0 \\ v_D &< 0 \\ v_E &< 0 \\ v_F &= 0 \\ v_G &> 0 \end{aligned}$$

ACCELERATION

★ Inst accelⁿ → Rate of change of velocity $\vec{a} = \frac{d\vec{v}}{dt}$

If $\vec{V} \rightarrow \text{constant}$

$$a = 0$$

Agar velocity badli to acceleration hai.



★ Average acceleration → $\frac{\text{change in velocity}}{\text{time}}$

$$= \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}$$

Q. A particle is moving on x-axis such that it's velocity is given at $V = t^2 + 4t + 10$.

Find acc at $t = 3$ sec

avg acc from $t = 0 \rightarrow t = 3$

Sol. $\vec{a} = \frac{dv}{dt} = 2t + 4$

At $t = 3$ $a = 2 \times 3 + 4 = 10$

Average acc = $\frac{V_f - V_i}{\text{time}}$

$t = 3, V_f = 9 + 12 + 10 = 31$

$t = 0, v_i = 10$

Average acc = $\frac{31 - 10}{3 - 0} = 7$

Q. $x = t^3 + 2t^2 + 5t$ Find velocity & acc at $t = 2$ sec

Sol. $v = \frac{dx}{dt} = 3t^2 + 4t + 5$

$a = \frac{dv}{dt} = 6t + 4$

$t = 2, a = 12 + 4 = 16$

$t = 2, v = 12 + 8 + 5 = 25$



Kaam ka Dabba

Ab tak ka final result ye hai ki:

★ x-t graph mein kisi point par tangent ka slope us point par velocity dega.

★ v-t graph mein kisi point par tangent ka slope us point par acc (acceleration) dega.

★ $\vec{v}, \vec{a} \Rightarrow \text{same sign} \Rightarrow \text{speed up}$

★ $\vec{v}, \vec{a} \Rightarrow \text{opposite sign} \Rightarrow \text{speed down.}$

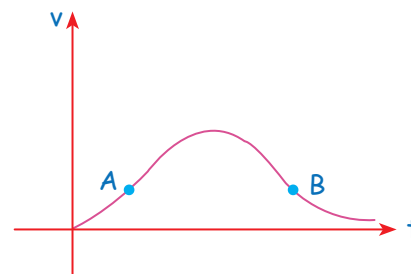
★ Average velocity = $\frac{\text{total displacement}}{\text{total time}}$

★ Average speed = $\frac{\text{total distance}}{\text{total time}}$

★ Average acc = $\frac{\text{change in velocity}}{\text{total time}} = \frac{\vec{v}_f - \vec{v}_i}{t}$

★ (a - t) graph ka slope kaddu dega

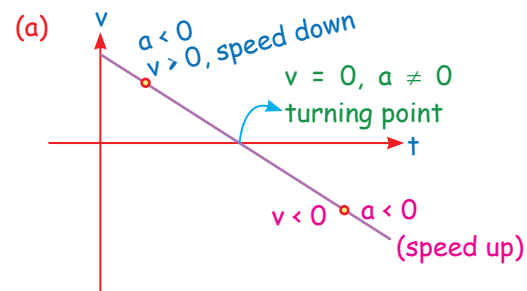
Q. Check if particle is slowing down or speed up at points A & B?



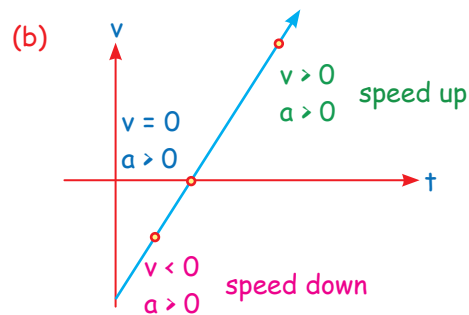
Sol. $v_A > 0, a_A > 0 \Rightarrow \text{speed up}$

$v_B > 0, a_B < 0 \Rightarrow \text{speed down}$

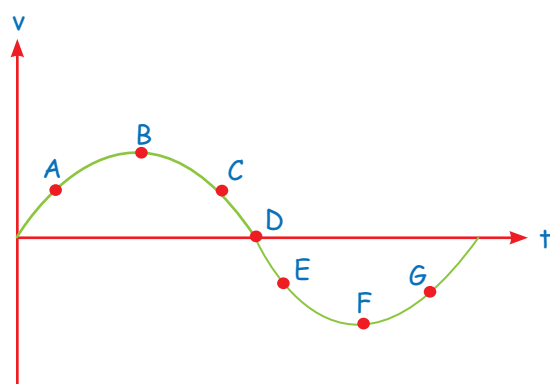
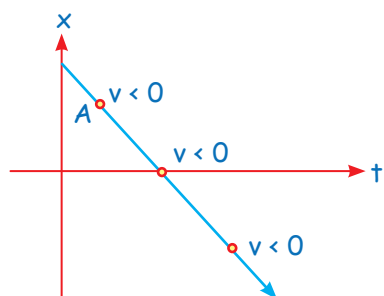
Q. Where is particle slowing down?



At turning point $\Rightarrow v = 0$ & v changes its sign at this point.



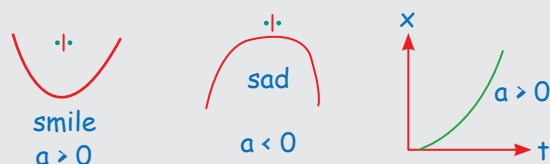
Q. Is velocity +ve or -ve in the x-t graph?



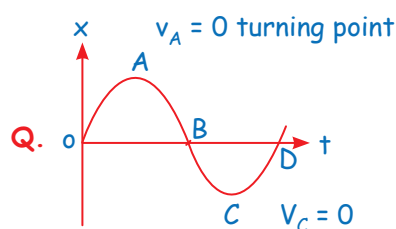
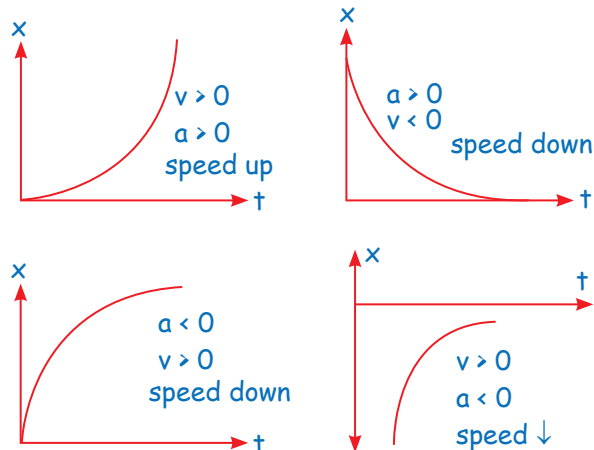
- $A \Rightarrow v > 0, a > 0$, speed up
 $B \Rightarrow v > 0, a = 0$
 $C \Rightarrow v > 0, a < 0$, speed down
 $D \Rightarrow v = 0, a < 0$
 $E \Rightarrow v < 0, a < 0$, speed up
 $F \Rightarrow v < 0, a = 0$
 $G \Rightarrow v < 0, a > 0$, speed down

1. Pehle ye dekho, given kya hai aur pucha kya hai?
2. x - y axis mai kya given hai.

x - t graph se direct acc ka sign pata karna ho.



Only for x - t graph



	v	a	Speed ↓↑
0 → A	+	-	Speed ↓
A → B	-	-	Speed up
B → C	-	+	Speed ↓
C → D	+	+	Speed ↑

INTEGRATION (MATHS)

Reverse of differentiation

$\frac{d}{dx} f(x) \Rightarrow$ differentiation of $f(x)$ wrt x

$\frac{dy}{dx} \Rightarrow$ differentiation of y wrt x

$\int y dx \Rightarrow$ integration of y wrt x

$\frac{d}{dx} f(x) = f'(x) \Rightarrow \int f'(x) = f(x) + c$, where c is constant of integration.

$f(x) \xrightarrow{\text{differentiation}} g(x)$
 $\downarrow$ integration
 $f(x) + c$

Indefinite Integration

$\int x^n dx = \text{Integration of } x^n \text{ wrt } x$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Physics mein sabse jyada yhi formula use hoga

Examples:

$$1. \int x^4 dx = \frac{x^{4+1}}{4+1} + C = \frac{x^5}{5} + C$$

$$2. \int x^7 dx = \frac{x^{7+1}}{7+1} + C = \frac{x^8}{8} + C$$

$$3. \int (x^2 + x^3) dx = \frac{x^3}{3} + \frac{x^4}{4} + C$$

$$4. \int 5x^4 dx = 5 \int x^4 dx \\ = 5 \cdot \frac{x^5}{5} + C = x^5 + C$$

$$5. \int (3x^2 + 7x^6) dx \\ \frac{3x^3}{3} + \frac{7x^7}{7} + C \Rightarrow x^3 + x^7 + C$$

$$6. \int x^{-3} dx = \frac{x^{-3+1}}{-3+1} + C \\ = \frac{-1}{2x^2} + C$$

$$7. \int \left(\frac{1}{x^9} + \frac{1}{x^{10}} \right) dx \\ = \int (x^{-9} + x^{-10}) dx \\ = \frac{x^{-9+1}}{-9+1} + \frac{x^{-10+1}}{-10+1} + C = -\frac{x^{-8}}{8} - \frac{x^{-9}}{9} + C$$

$$8. \int dx = \int 1 dx \\ = \int x^0 dx \\ = \frac{x^{0+1}}{0+1} + C = x + C$$

$$9. \int 5 dx = 5 \int dx = 5x + C$$

$$10. \int (2t + 7t^6 + 10) dt = \frac{2t^2}{2} + \frac{7t^7}{7} + 10t + c \\ = t^2 + t^7 + 10t + c$$

FORMULA

$$\star \int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\star \int e^x dx = e^x + c$$

$$\star \int \frac{1}{x} dx = \ln x + c$$

$$\star \int \sin x = -\cos x + c$$

$$\star \int \cos x dx = \sin x + c$$

$$\star \int \frac{1}{ax+b} dx = \frac{1}{a} \ln(ax+b) + c$$

$$\star \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + c$$

$$\star \int \cos(ax+b) dx = +\frac{1}{a} \sin(ax+b) + c$$

$$\star \int 2 \cos(2x+3) = \sin(2x+3)$$

$$\star \int \cos(2x+3) = \frac{1}{2} \sin(2x+3) + c$$

$$\star \int \frac{1}{4x+5} dx = \frac{1}{4} \ln(4x+5) + c$$

Definite Integration

$$\begin{array}{c} \text{Upper} \\ \text{limit of } x \\ \uparrow \\ x_f = b \\ \int f(x) \cdot dx = g(x) \quad \left| \begin{array}{l} x_f = b \\ x_i = a \end{array} \right. = g(b) - g(a) \\ \downarrow \\ x_i = a \\ \text{Lower} \\ \text{limit of } x \end{array}$$

Ex (1):

$$\int_0^3 x^2 dx = \frac{x^3}{3} \Big|_0^3 = \left(\frac{3^3}{3} \right) - \left(\frac{0^3}{3} \right) = 9 \text{ Ans}$$

Ex (2):

$$\int_2^3 4x^3 dx = 4 \frac{x^4}{4} \Big|_2^3 = x^4 \Big|_2^3 = 3^4 - 2^4 = 81 - 16 = 65 \text{ Ans}$$

Ex (3):

$$\int_0^{\pi/2} \cos x dx = \sin x \Big|_0^{\pi/2} = \left(\sin \frac{\pi}{2} \right) - \sin 0 = 1 - 0 = 1$$

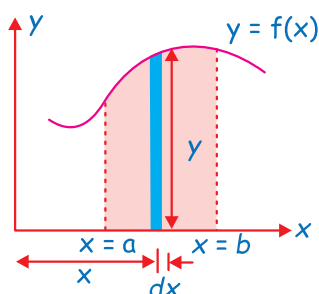
Ex (4):

$$\int_2^3 e^x dx = e^x \Big|_2^3 = e^3 - e^2$$

$$\begin{aligned} \text{Q. } \int_0^1 (3x^2 + 2x) &= \left[x \frac{x^3}{3} + 2 \frac{x^2}{2} \right]_0^1 \\ &= (x^3 + x^2) \Big|_0^1 = (1^3 + 1^2) - (0^3 + 0^2) = 2 \text{ Ans} \end{aligned}$$

APPLICATION OF INTEGRATION

★ Area under the curve :



Area of small element = $y dx = f(x) dx$

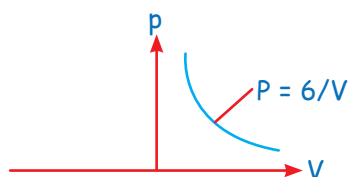
If we sum up all areas between $x = a$ and $x = b$ then

$$\int_a^b f(x) dx = \text{shaded area between curve and x-axis.}$$

★ Average value

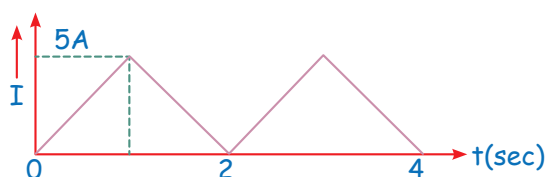
$$\begin{aligned} \text{if } y = f(x) \text{ then } \bar{y} \text{ or } y_{\text{average}} &= \frac{\int_{x=a}^{x=b} f(x) dx}{b-a} \\ &= \frac{\text{Area under the curve}}{\text{interval}} \end{aligned}$$

Q. A gas expands its volume from V to $3V$ as shown in figure. Calculate the work in this process if $W = \int p dv$.



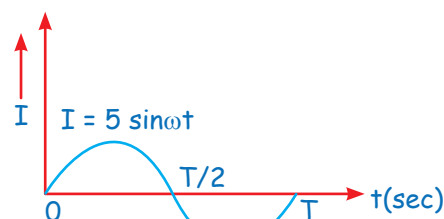
$$\text{Sol. } W = \int_V^{3V} p dv \Rightarrow \int_V^{3V} \frac{6}{v} dv = [6 \log_e v]_V^{3V} = 6 \ln 3$$

Q. Calculate average value of current from $t = 0$ to $t = 4$ seconds.



$$\text{Sol. } I_{\text{average}} = \frac{\int_a^b I dt}{\int_a^b dt} = \frac{\text{total area}}{\text{total time}} = \frac{5 \times 2}{4} = 2.5 \text{ amp}$$

Q. Calculate average value of current from $t = 0$ to $t = T$ seconds ($T = 2\pi/\omega$).



$$\begin{aligned} \text{Sol. } I_{\text{average}} &= \frac{\int_0^T 5 \sin \omega t}{\int_0^T dt} = \frac{5}{\omega T} [-\cos \omega t]_0^T \\ &= -\frac{5}{\omega T} [\cos \omega T - \cos 0] = 0 \quad \left[\omega = \frac{2\pi}{T} \right] \end{aligned}$$

★ For any moving object, the average speed can never be zero or negative, as total distance covered is always +ve.

★ If a particle travels distances $s_1, s_2, s_3, \dots$, etc., at different speeds $v_1, v_2, v_3, \dots$, etc., respectively, then

$$v_{\text{av}} = \frac{\Delta s}{\Delta t} = \frac{\sum s_i}{\sum (s_i / v_i)}$$

If $s_1 = s_2 = \dots = s_n = s$,

$$\text{Then } \frac{1}{v_{\text{av}}} = \frac{1}{n} \left[\frac{1}{v_1} + \frac{1}{v_2} + \dots \right] = \frac{1}{n} \sum \frac{1}{v_i}$$

Special case: If a particle moves a distance at speed v_1 and comes back to initial position with speed v_2 , then

$$v_{\text{av}} = \frac{2v_1 v_2}{v_1 + v_2}$$

★ If a particle travels at speeds $v_1, v_2, \dots$, etc., for time intervals $t_1, t_2, \dots$, then

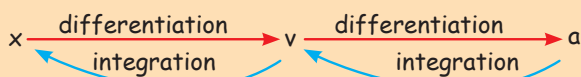
$$v_{\text{av}} = \frac{\Delta s}{\Delta t} = \frac{v_1 t_1 + v_2 t_2 + \dots + v_n t_n}{t_1 + t_2 + \dots + t_n} = \frac{\sum v_i t_i}{\sum t_i}$$

Special case: If a particle moves for two equal intervals of time at different speeds, then

$$v_{\text{av}} = \frac{v_1 + v_2}{2}$$



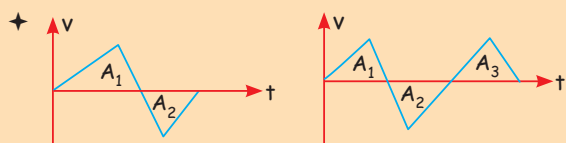
Kaam ka Dabba



$\int y \cdot dx = \text{Area under curve}$
curve aur x-axis k bich ka Area

- ✦ a-t graph ka area change in velocity deta hai
- ✦ v-t graph ka area change in position deta hai
- bole toh

v-t graph ka area displacement deta hai



$$\text{Displacement} = A_1 - A_2$$

$$\text{Distance} = A_1 + A_2$$

$$\text{Displacement} = A_1 - A_2 + A_3$$

$$\text{Distance} = A_1 + A_2 + A_3$$

- ✦ (x - t) graph ka slope $\Rightarrow v$
- ✦ (v - t) graph ka slope $\Rightarrow a$
- ✦ $v = \frac{dx}{dt}$, $a = \frac{dv}{dt}$
- ✦ Displacement = $\int v dt = \text{Area}$
- ✦ change in velocity = $\int a dt$
- ✦ St line $\rightarrow$ slope $\rightarrow$ const
- ✦ (v - t) is st. line $\rightarrow a = \text{constant}$
- ✦ If (x-t) is straight line $\rightarrow v = \text{constant}$ and $a = 0$.
- ✦ If $a = 0 \rightarrow v = \text{constant} \rightarrow (x-t)$ straight line
- ✦ If $a = \text{const} \rightarrow (v-t)$ straight line $\rightarrow (x-t)$ parabola
- ✦ If a body moves with uniform acceleration and velocity changes from u to v in a time interval, then average velocity = $\frac{v+u}{2}$.
- ✦ If a body moving with uniform acceleration has velocities u and v at two points in its path, then the velocity at the midpoint of given two points = $\sqrt{\frac{u^2 + v^2}{2}}$.

Q. At $t = 0$, particle is at $x = 10$ particle is moving such that its velocity vs time relation is given as $V = 3t^2 + 2t$. Find location of particle, $t = 1$ and $x = f(t)$

Sol. $x = \int v dt = \int (3t^2 + 2t) dt = t^3 + t^2 + c$

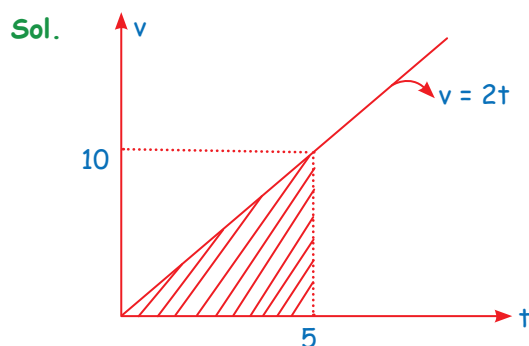
at $t = 0$, $x = 10$ (put the value)

$$x = t^3 + t^2 + c$$

$$10 = 0 + 0 + c \Rightarrow c = 10$$

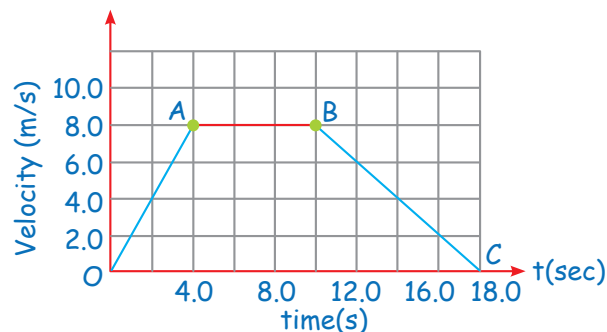
$$x = t^3 + t^2 + 10 \text{ (now you can find } x \text{ for any } t)$$

Q. A particle is moving along x-axis. Its velocity vs time relation is given as $v = 2t$. Find displacement from $t = 0$, $t = 5$



$$\text{Area} = \frac{1}{2} \times 5 \times 10 = 25 = \text{displacement}$$

Q. What is the acceleration for each graph segment in figure? Describe the motion of the object over the total time interval. Also calculate displacement.



Sol. Segment OA; $a = \frac{8-0}{4-0} = 2 \text{ m/s}^2$

Segment AB; graph horizontal i.e., slope zero i.e., $a = 0$

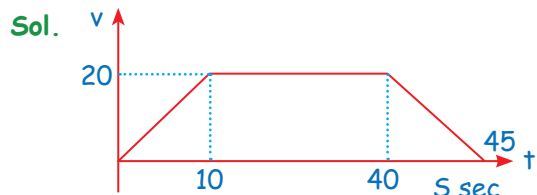
$$\text{Segment BC; } a = \frac{0-8}{18-10} = -1 \text{ m/s}^2$$

The graph is trapezium. Its area between $t = 0$ to $t = 18$ s is displacement.

$$\text{Area of } v-t \text{ graph} = \text{displacement} = \frac{1}{2} (18 + 6) \times 8 = 96 \text{ m}$$

Particle accelerates uniformly for first 4 sec., then moves with uniform velocity for next 6 sec. and then retards uniformly to come to rest in next 8 sec.

- Q.** A particle starts from rest, accelerates at 2 m/s^2 for 10 s & then goes at constant velocity for 30s and then deaccelerate at 4 m/s^2 till it stops. What is the distance travelled by it.



$$\text{Area} = \frac{1}{2} \times (30 + 45) \times 20 = 750$$

$$\left. \begin{aligned} v &= u + at \\ s &= ut + \frac{1}{2}at^2 \\ v^2 &= u^2 + 2as \end{aligned} \right\} \begin{aligned} a &= \text{const} \\ u &= \text{initial velocity} \\ v &= \text{final velocity} \\ a &= \text{acc} \\ s &= \text{displacement} \end{aligned}$$



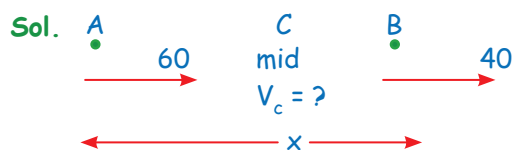
Bhai, ye equation tabhi lagana jab a (acceleration) constant ho.

- Q.** A particle start motion having initial velocity 20 m/s and it move with const acc of 10 m/s^2
1. Find velocity at $t = 4 \text{ sec}$
 2. Find displacement of particle from $t = 0$ to $t = 4 \text{ sec}$
 3. Find displacement in $4^{\text{th}} \text{ sec}$

Sol. 1. $t = 4, v = u + at$
 $= 20 + 10 \times 4$
 $v = 60$

2. $s = ut + \frac{1}{2}at^2$
 $= 20 \times 4 + \frac{1}{2} \times 10 \times 4^2 = 160$

- Q.** A truck travelling with uniform acceleration crosses two points A & B with velocities 60 m/s & 40 m/s respectively. The speed of the body at the midpoint of A & B is nearest to.



$A \rightarrow B \quad 40^2 = 60^2 + 2ax \Rightarrow 2ax = -2000 \dots (i)$

$A \rightarrow C \quad V_c^2 = 60^2 + 2a \times \frac{x}{2} \dots (ii)$

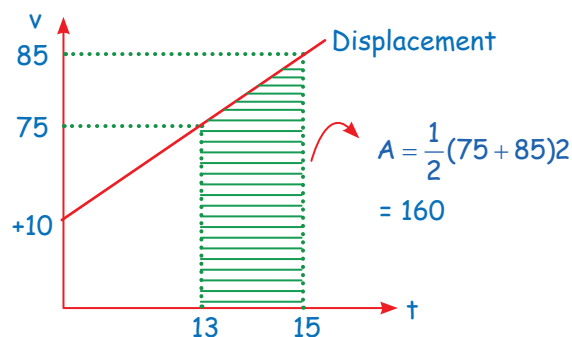
from (i) & (ii)

$$V_c^2 = 3600 - 1000 = 2600$$

$$V_c = 10\sqrt{26}$$

- Q.** A particle having initial velocity 10 m/s move with a constant acceleration 5 ms^{-2} , for a time 15 second along a straight line, what is the displacement of the particle in last 2 second?

Sol.



- Q.** A bullet is moving with a velocity of 200 cm/s . penetrates a wooden block & comes to rest after travelling 4 cm insides it. What velocity is needed for travelling distance of 9 cm in same block?

Sol. $0^2 = 2^2 + 2a \times \frac{4}{100} \quad (v^2 = u^2 + 2as)$

$$-4 = \frac{8}{100}a \Rightarrow a = -50 \text{ m/s}^2$$

To penetrate 9 cm , let initial velocity be 'u'.

$$0^2 = u^2 + 2as$$

$$0 = u^2 + 2 \times (-50) \times \frac{9}{100}$$

$$u^2 = 9$$

$$u = 3 \text{ m/s} = 300 \text{ cm/s}$$

- Q.** A bullet going with speed 350 m/s enters a concrete wall and penetrates a distance of 5.0 cm before coming to rest. Find the deacceleration

Sol. $U_f = 0, a = ?$

$$0^2 = (350)^2 + 2 \times a \times \frac{5}{100}$$

$$a = -(350)^2 \times 10$$

$$a = -1225000 \text{ m/s}^2$$

Q. A particle covered 100 m distance in first 10 s of its journey and in next 10 s it travel 200 m. Find distance travelled in next 10s? (acc is const)

Sol. $t = 0 \rightarrow t = 10$ $100 = u \times 10 + \frac{1}{2}a \times 10^2$... (i)

$t = 0 \rightarrow t = 20$ $300 = u \times 20 + \frac{1}{2}a(20)^2$... (ii)

Solve (i) and (ii) and get $a = 1$, $u = 5$

Let it travel x_3 in last 10s so in 30s a travel $300 + x_3$

$$300 + x_3 = 5 \times 30 + \frac{1}{2} \times 1 \times (30)^2$$

Solve and get $x_3 = 300$

Q. A particle moving with initial velocity of 10 m/s towards East has an acceleration of 5 m/s² towards west. Find the displacement and distance travelled by the particle in first 4 seconds?

Sol. $u = 10 \text{ m/s}$
 $a = -5 \text{ m/s}^2$ $t = 2$

$$v = u + at \Rightarrow 0 = 10 - 5t \Rightarrow t = 2 \text{ s}$$

The direction of velocity changes after two seconds.

$$S = 10 \times 2 + \frac{1}{2}(-5) \times 2^2 = 0 = \text{displacement}$$

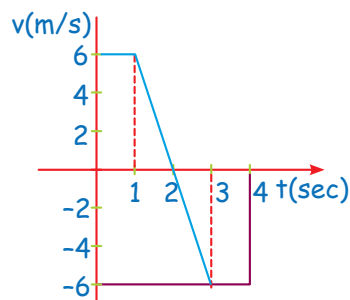
Distance travelled is not equal to displacement because during course of journey, velocity changes direction.

$$D = S(\text{at } 2\text{s}) + |S(\text{at } 4\text{s}) - S(\text{at } 2\text{s})|$$

$$= \left(10 \times 2 - \frac{1}{2} \times 5 \times 2^2 \right) + \left| 0 - (10 \times 2) - \frac{1}{2} \times 5 \times 2^2 \right|$$

$$= 10 + 10 = 20 \text{ m}$$

Q. A particle moves along a straight line along x-axis. At time $t = 0$, its position is at $x = 0$. The velocity v m/s of the object changes as a function of time t seconds as shown in the figure.



(i) What is x at $t = 1 \text{ sec}$?

(ii) What is the acceleration at $t = 2 \text{ sec}$?

(iii) What is x at $t = 4 \text{ sec}$?

(iv) What is the average speed between $t = 0$ and $t = 3 \text{ sec}$?

Sol. (i) x is displacement at $t = 1 \text{ sec}$.

Area under the $v-t$ curve gives displacement

From $t = 0$ to $t = 1 \text{ sec}$.

$$x = 6 \times 1 = 6 \text{ m}$$

(ii) Slope of the $v-t$ curve gives acceleration from the given $v-t$ curve

Slope at $t = 2 \text{ sec}$. gives acceleration at $t = 2 \text{ sec}$.

$$\tan \theta = a = -\frac{6}{1} = -6 \text{ m/s}^2$$

(iii) x (at $t = 4 \text{ sec}$):

Area under the curve from $t = 0$ to $t = 4 \text{ sec}$

$$= 6 \times 1 + \frac{1}{2} \times 6 \times 1 - \frac{1}{2} \times 6 \times 1 - 6 \times 1 = 0$$

$$\Rightarrow x(t = 4) = 0 \text{ m}$$

(iv) Average speed from $t = 0$ to $t = 3 \text{ sec}$.

Displacement from $t = 0$ to $t = 2 \text{ sec}$. = Area

$$\text{under the curve} = 6 + \frac{1}{2} \times 6 \times 1 = 9 \text{ m}$$

Displacement from $t = 2$ to $t = 3 \text{ sec}$.

$$= -\frac{1}{2} \times 6 \times 1 = -3 \text{ m}$$

$$\text{Distance from } t = 0 \text{ to } t = 3 \text{ sec} = |9| + |-3| = 12 \text{ m}$$

$$\text{Average speed} = \frac{\text{Distance}}{\text{Time}} = \frac{12}{3} = 4 \text{ m/s}$$

Question Practice on Variable Acceleration and Utptang Integration

Q. The acceleration a of a particle moving in one dimension is given by $a = 6 - 2t$. If the particle is initially at $x = 0$ and its velocity is 2 m/s, find its position and velocity at time t .

Sol. $\frac{dv}{dt} = 6 - 2t$

$$\int_2^v dv = \int_0^t (6 - 2t) dt$$

$$\Rightarrow v - 2 = (6t - t^2)|_0^t = 6t - t^2 \Rightarrow v(t) = 2 + 6t - t^2$$

To find position, we integrate velocity.

$$v = \frac{dx}{dt} = 2 + 6t - t^2 \Rightarrow dx = (2 + 6t - t^2) dt$$

$$\int_0^x dx = \int_0^t (2 + 6t - t^2) dt = 2t + 3t^2 - \frac{t^3}{3}$$

$$\text{or } x(t) = 2t + 3t^2 - \frac{t^3}{3}$$

- Q.** The retardation of a car when its engine is shut off depends on its velocity as $a = -\alpha v$ where α is positive constant. Find the total distance travelled by the car if its initial velocity is 20 m/s and $\alpha = 0.5/\text{s}$.

Sol. $\frac{dv}{dt} = -\alpha v$

$$\frac{dv}{dx} \left(\frac{dx}{dt} \right) = -\alpha v \Rightarrow \frac{v dv}{dx} = -\alpha v$$

$$\text{or } dv = -\alpha dx$$

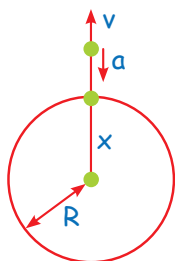
$$\int_{20}^0 dv = -\alpha \int_0^d dx \Rightarrow v|_{20}^0 = -\alpha x|_0^d$$

$$-20 = -\alpha d$$

$$d = \frac{20}{\alpha} = \frac{20}{0.5} = 40 \text{ m}$$

- Q.** With what velocity in vertical upward direction should a body be projected from the surface of earth so that it reaches a height equal to radius of earth? The acceleration of body is given by $a = -\frac{GM}{x^2}$ where x is the distance from centre of earth and M is the mass of earth.

- Sol.** Note that acceleration due to gravity is nearly constant near the surface of earth. But if the height become too large its dependence on distance can not be ignored.



$$a = \frac{dv}{dt} = -\frac{GM}{x^2}$$

$$\text{or } \frac{dv}{dx} \cdot \frac{dx}{dt} = -\frac{GM}{x^2} \Rightarrow v dv = -\frac{GM}{x^2} dx$$

At the highest point, velocity is zero. Also note $x_i = R$ and $x_f = 2R$.

$$\int_u^0 v dv = -GM \int_R^{2R} \frac{dx}{x^2}$$

$$\frac{v^2}{2} \Big|_u^0 = -GM \int_R^{2R} x^{-2} dx = \frac{GM}{x} \Big|_R^{2R}$$

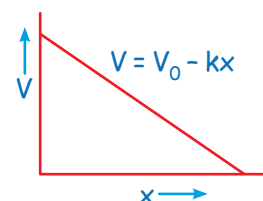
$$\Rightarrow -\frac{u^2}{2} = GM \left[\frac{1}{2R} - \frac{1}{R} \right]$$

$$\Rightarrow u^2 = \frac{GM}{R} \Rightarrow u = \sqrt{\frac{GM}{R}} = \sqrt{\frac{GM}{R^2} R}$$

$$\therefore g = \left(\frac{GM}{R^2} \right)$$

$$\therefore u = \sqrt{gR} = 8 \text{ km/s } [\because R = 6400 \text{ km}, g = 10 \text{ m/s}^2]$$

- Q.** A particle is moving along x-axis with velocity V which varies according to the law $V = V_0 - Kx$ here V_0 and K are constants. Choose the correct acceleration vs time plot for the time interval when particle moves from $x = 0$ to $x = \frac{V_0}{K}$.



Sol. $V = V_0 - Kx$

$$\frac{dx}{dt} = (V_0 - Kx) \Rightarrow \int_0^x \frac{dx}{(V_0 - Kx)} = \int_0^t dt$$

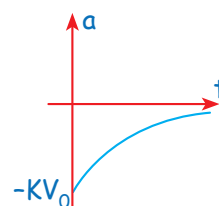
$$x = \frac{V_0}{K} (1 - e^{-Kt})$$

$$\therefore \frac{dx}{dt} = + V_0 e^{-Kt} \Rightarrow a = \frac{d^2x}{dt^2} = -KV_0 e^{-Kt}$$

$$\text{At } t = 0, a = -KV_0$$

$$\text{At } t = \infty, a = 0$$

Therefore, graph is as shown



- Q.** A block of mass m is fired horizontally along a level surface that is lubricated with oil. The oil provides a viscous resistance that varies as the $3/2$ power of the speed. If the initial speed of the block is v_0 at $x = 0$, find the maximum distance reached by the block. Assume no resistance to motion other than that provided by the oil.

Sol. $F = -v^{3/2}$

$a = -\frac{1}{m}v^{3/2}$ [$\because F = ma$]

$v \frac{dv}{dx} = -\frac{1}{m}v^{3/2}$

$\int_{v_0}^0 v^{-1/2} dv = -\frac{1}{m} \int_0^d dx$

$2mv_0^{1/2} = d$

Q. Acceleration of particle moving rectilinearly is $a = 4 - 2x$ (where x is position in metre and a in m/s^2). It is at rest at $x = 0$. At what position x (in metre) will the particle again come to instantaneous rest?

Sol. $\frac{v dv}{dx} = 4 - 2x$

$\int_0^v v dv = \int_0^x (4 - 2x) dx \Rightarrow \frac{v^2}{2} = 4x - x^2$

when $v = 0$, $4x - x^2 = 0$

$x = 0, 4$

$\therefore$ At $x = 4$ m, the particle will again come to rest.

Question for Practice

Q. A particle start motion at $t = 0$ from $x = +10$, such that its ' v ' vs t relation is given as $\bar{v} = 4t^3 + 3t^2 + 2t$. Find location of particle at $t = 1$ sec.

Sol. $x = \int v dt$

$x = t^4 + t^3 + t^2 + c$

$t = 0, x = 10$ (put $t = 0, x = 10$)

$10 = 0 + 0 + 0 + c$

$c = 10$

$x = t^4 + t^3 + t^2 + 10$

$t = 1, x = 13$

Q. A particle start motion from $x = 5$, at $t = 0$ such that $v = t^2 + t$. Find location of particle at $t = 6$ sec.

Sol. $dx = v dt$

$\int_{x=5}^{x_f} dx = \int_{t=0}^{t=6} (t^2 + t) dt$

$x|_5^{x_f} = (t^3/3 + t^2/2)|_0^6$

$x_f - 5 = \left(\frac{6^3}{3} + \frac{6^2}{2}\right) - \left(\frac{0^3}{3} + \frac{0^2}{2}\right)$

$x_f - 5 = 72 + 18 - 0$

$x_f = 90 + 5$

$x_f = 95$ Ans

Q. A particle start motion from rest from origin at $t = 0$ such that $a = 6t$ find v, x (location) at, $t = 2$ sec

Sol. $a = 6t$

$v = \int a dt = \int 6t dt$

$v = 3t^2 + c$

$v = 3t^2 + c$

$t = 0, v = 0$

$0 = 0 + c$

$c = 0$

$v = 3t^2$

$t = 2, v = 12$

$x = \int 3t^2 dt = 3 \frac{t^3}{3} + c'$

$x = t^3 + c'$

$t = 0, x = 0, 0 = 0 + c' \rightarrow c' = 0$

$x = t^3$

$t = 2, x = 8, v = 12$

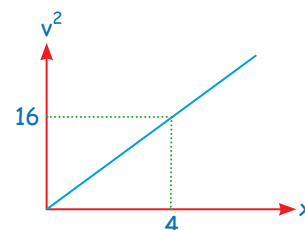
Q. If $v = x^2 + 3x$ Find acc at $x = 2$

Sol. $a = v \frac{dv}{dx} \quad \frac{dv}{dx} = 2x + 3$

$a = (x^2 + 3x)(2x + 3)$

put $x = 2, a = 70$

Q. Find acc at $x = 4$

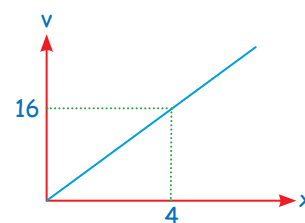


Sol. slope = $\frac{dy}{dx} = \frac{d(v^2)}{dx}$

$16/4 = 2v \frac{dv}{dx}$

$4 = 2a \rightarrow a = 2$

Q. Find acc at $x = 4$



Sol. $\text{slope} = \frac{dy}{dx} = \frac{dv}{dx} = 16/4$

$$a = v \frac{dv}{dx}$$

$$a = 16 \times \frac{16}{4} = \boxed{64}$$

Q. $x = \sqrt{v+1}$ find acc at $x = 5$

Sol. $x^2 = v + 1$

$$v = x^2 - 1$$

$$a = v \frac{dv}{dx} = (x^2 - 1)(2x)$$

$$x = 5 \Rightarrow a = 240$$

Q. Acceleration of a particle moving on x-axis having initial speed v_0 with distance from origin is given by $a = \sqrt{x}$ Distance covered by particle where its speed become thrice that of initial speed.

Sol. $a = \sqrt{x}$ $a > 0$, speed 1, $v > 0$

$$v \frac{dv}{dx} = \sqrt{x}$$

$$\int v dv = \int \sqrt{x} dx$$

$$\int_{v_0}^{3v_0} v dv = \int_0^{x_f} x^{1/2} dx$$

$$\frac{v^2}{2} \Big|_{v_0}^{3v_0} = \frac{x^{1/2+1}}{1/2+1} \Big|_0^{x_f}$$

$$\frac{9v_0^2}{2} - \frac{v_0^2}{2} = \frac{2}{3} x^{3/2} \Big|_0^{x_f} = \frac{2}{3} (x_f)^{3/2}$$

$$\frac{8v_0^2}{2} = \frac{2}{3} (x_f)^{3/2} \Rightarrow x_f = \left(6v_0^2\right)^{2/3}$$

Q. $x = a \sin \omega t$

$$y = a(1 - \cos \omega t) = a - a \cos \omega t$$

Find $\vec{v}$, speed of particle

Sol. $v_x = a\omega \cos \omega t$

$$v_y = 0 - a\omega [-\sin \omega t]$$

$$v_y = a\omega \sin \omega t$$

$$\vec{v} = (a\omega \cos \omega t) \hat{i} + (a\omega \sin \omega t) \hat{j}$$

speed = ? magnitude

$$|\vec{v}| = \sqrt{(a\omega \cos \omega t)^2 + (a\omega \sin \omega t)^2}$$

$$= a\omega \sqrt{\cos^2 \omega t + \sin^2 \omega t}$$

$$\text{speed} = a\omega$$

MOTION UNDER GRAVITY

It's very important article jisme ham particle ka motion under the effect of gravity padhenge.....

Assumption

1. Air resistance force is neglected until mention.
2. Variation of gravity g is neglected until mention.

Bhaiya, kinematics ke ques mein samajh nahi aata kaise karna hai. Sign convention mein bhi dikkat hoti hai. solution toh samajh aa jate hain par sawal khud se nahi bante. Iska koi ilaaj...



Haan bhai, sabse pehle toh zor se ghutna lagao... Fir jo given hai woh likh lo, jo poocha hai woh bhi likh lo aur dekho kaun-si eqn of motion (equation of motion) connect ho rahi hai.

+/- jahan man kare ooper (up) ya neeche (down) maan sakte ho, answer same aayega. Bas is baat ka dhyaan rakhna, u , s , a ko with sign likhna hai.

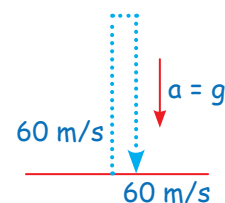
Q. A particle is projected from ground vertically upward with velocity 60 m/s. (1) Find velocity and location of particle at $t = 10$ (2). When will particle come to rest (3). Find $h_{\max}$, time of flight.

Sol. Sign convention

Let upward direction (+ve)

$$u = 60 \text{ upar} = +60$$

$$a = 10 \text{ neeche} = -10$$



$$t = 10, v = u + at = 60 + (-10)(10)$$

$$v = \ominus 40$$

niche

$$\text{location} = y = 60 \times 10 + \frac{1}{2} (-10) (10)^2 = \oplus 100$$

upar

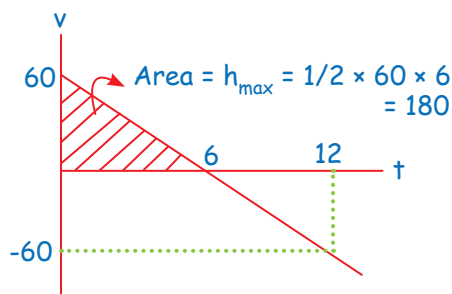
When particle will come to rest

$$v = u + at \Rightarrow 0 = 60 - 10t \Rightarrow t = 6$$

$$0^2 = 60^2 + 2 \times (-10) h_{\max}$$

$$h_{\max} = 180$$

✦ Time of flight = 6 + 6 = 12 sec



Bhaiya jab particle upar jaata hai tab uska acceleration neeche hota hai aur jab neeche aata hai tab bhi uska acceleration neeche hota hai.



Haan bhai, abhi ke liye bas tu yeh yaad rakh agar particle hawa mein hai toh uska acceleration (g) hoga chaahe upar jaaye ya neeche. particle udhar jaata hai jis taraf uski velocity hoti hai aur uska acceleration udhar hota hai jis taraf uspe net force hota hai.



Q. A particle is thrown in vertically downward with velocity 60 m/s from top of a tower of height 320 m. Find when particle will hit the ground & with what velocity & $V_f = ?$

Sol. Let (Downward +ve)

$$u = +60 \quad a = +10$$

✦ displacement $s = ut + \frac{1}{2}at^2$

$$+320 = 60t + \frac{1}{2} \times 10 \times t^2$$

$$t^2 + 12t - 64 = 0$$

$$(t + 16)(t - 4) = 0 \Rightarrow t = 4s$$

✦ $V_f = ?$

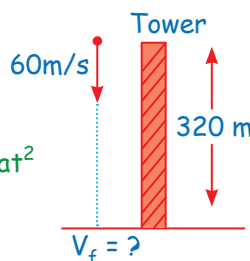
$$V^2 = u^2 + 2as$$

$$V^2 = 60^2 + 2 \times 10 \times 320$$

$$V^2 = 3600 + 6400$$

$$V^2 = 10000$$

$$V_f = \text{velocity } V = 100 \text{ (niche)}$$



Q. A particle is thrown in vertically upward with velocity 40 m/s from top of a tower of height 240 m. Find when particle will hit the ground & with what velocity & $V_f = ?$

Sol. (1) let (up = +ve)

$$u = +40, a = -10, s = -240$$

$$s = ut + \frac{1}{2}at^2$$

$$-240 = 40t + \frac{1}{2}(-10)t^2$$

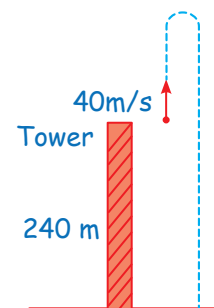
$$5t^2 - 40t - 240 = 0$$

$$(t - 12)(t + 4) = 0$$

$$t = 12$$

$$(2) V^2 = u^2 + 2as = (40)^2 + 2 \times (-10) \times (-240)$$

$$V = 80$$



Q. A balloon is rising upward with constant velocity 40 m/s. When it reaches at a height of 240 m from ground a particle is dropped from it. Find when will particle hit the ground with what velocity?

Ans. $t = 12, v = 80 \text{ m/s}$

SKC

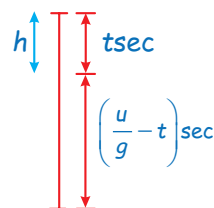


Agar hum kisi chalti hui gaadi, lift aur udte hue gubbara (balloon) se kisi particle ko drop karein, toh drop ke just baad particle ki velocity wahi hoti hai jo us gaadi, lift aur gubbara (balloon) ki ho. So, abe, ye sawal toh wahi tower wala sawal hai.

Q. If a ball is thrown vertically upwards with speed u , the distance covered during the last t seconds of its ascent is

Sol. If ball is thrown with velocity u , then time of

$$\text{ascent} = \frac{u}{g}$$



$$\text{Velocity after } \left(\frac{u}{g} - t\right) \text{ sec, } v = u - g\left(\frac{u}{g} - t\right) = gt.$$

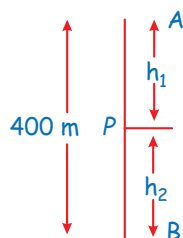
$$\text{So, distance in last 't' sec, } 0^2 = (gt)^2 - 2(g)h.$$

$$h = \frac{1}{2}gt^2.$$

Q. A man drops a ball downside from the roof of a tower of height 400 m. At the same time another ball is thrown upside with a velocity 50 m/s from the foot of tower. What is the height from the foot of the tower where the two balls would meet?

Sol. Let both balls meet at point P after time t .
The distance travelled by ball A

$$h_1 = \frac{1}{2}gt^2 \quad \dots(i)$$



The distance travelled by ball B

$$h_2 = ut - \frac{1}{2}gt^2 \quad \dots(ii)$$

By adding (i) and (ii) $h_1 + h_2 = ut = 400$

(Given $h = h_1 + h_2 = 400$)

$$\therefore t = 400/50 = 8 \text{ s and } h_1 = 320 \text{ m, } h_2 = 80 \text{ m}$$

Q. Water drops fall at regular intervals from a tap which is 5 m above the ground. The third drop is leaving the tap at the instant the first drop touches the ground. How far above the ground is the second drop at that instant?

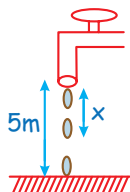
Sol. Let the interval between each drop be t then from question

$$\text{For first drop } \frac{1}{2}g(2t)^2 = 5 \quad \dots(i)$$

$$\text{For second drop } x = \frac{1}{2}gt^2 \quad \dots(ii)$$

By solving (i) and (ii) $x = \frac{5}{4}$ and

$$\text{Hence required height } h = 5 - \frac{5}{4} = 3.75 \text{ m.}$$



Q. A balloon is at a height of 81 m and is ascending vertically upward with a velocity of 12 m/sec. A body of 2 kg weight is dropped from it. If $g = 10 \text{ m/s}^2$ the body will reach the surface of the earth in

Sol. As the balloon is going up so initial velocity of balloon

$$= +12 \text{ m/s,}$$

$$\Delta y = -81 \text{ m; } a = -g = -10 \text{ m/s}^2$$

$$\text{By applying } h = ut + \frac{1}{2}gt^2; -81 = 12t - \frac{1}{2}(10)t^2$$

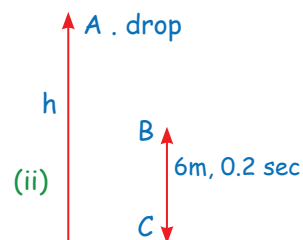
$$\Rightarrow 5t^2 - 12t - 81 = 0$$

$$\Rightarrow t = \frac{12 \pm \sqrt{144 + 1620}}{10} = \frac{12 \pm \sqrt{1764}}{10} = 5.4 \text{ s}$$

Q. A ball dropped from a height h from ground, if it take 0.2 sec to cross the last 6m before hitting the ground. Find height from which it was dropped.

$$\text{Sol. } t_{A \rightarrow C} = t = \sqrt{\frac{2h}{g}} \quad \dots(i)$$

$$t_{A \rightarrow B} = t - 0.2 = \sqrt{\frac{2(h-6)}{g}} \quad (ii)$$



PROJECTILE MOTION

Bhai, ye bahut mazedar chapter hai. Bas yahan x aur y dono mein motion ho raha hai. Bas ye yaad rakhna, x aur y do independent axis hain, toh dono ke saath independently khelna hai. Jab x ke saath ho toh y ko bhul jao aur jab y ke saath ho toh x ko bhul jao. It's like your gf... Abe, samajh gaya na?



In this chapter

★ Air Resistance neglected until mention

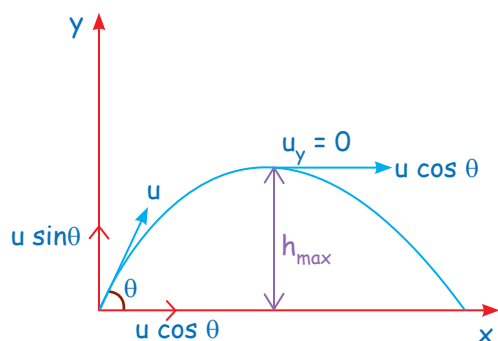
★ variation of g neglected $g = 10 \text{ m/s}^2$ (downward)

$$g = 9.8 \text{ m/s}^2$$

★ Since, $a_x = 0 \Rightarrow v_x = \text{const} = u \cos \theta$

★ Particle chahe upar jaa raha ho ya neeche aa raha ho, uska acceleration neeche ki taraf g hoga.

A particle is projected with speed u at angle θ with the horizontal as shown in fig.



1. Time of flight (T)

$$u_y = u \sin \theta,$$

$$a_y = -g$$

$$v = u + at$$

$$0 = u \sin \theta - gt$$

$$t = u \sin \theta / g \text{ (time to reach highest point)}$$

$$T = 2t = \frac{2u \sin \theta}{g}$$

2. Maximum height ($h_{\max}$)

$$v^2 = u^2 + 2as \quad (y)$$

$$0 = (u \sin \theta)^2 - 2g h_{\max}$$

$$h_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$

Here, Kinematics ka sabse bada hathiyar $a_x = 0$ means horizontal me velocity constant rahegi.

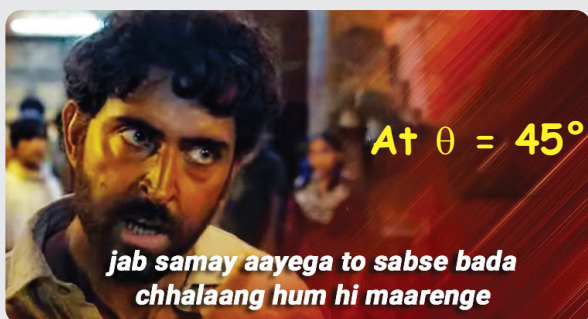
3. Range = $u \cos \theta \times T$

$$R = u \cos \theta \times \frac{2u \sin \theta}{g} = \frac{u^2 2 \sin \theta \cos \theta}{g}$$

$$R = \frac{u^2 \sin 2\theta}{g}$$

For R to be maximum by keeping u fix $\sin 2\theta$ should be maximum $\Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$

If u is fixed then from ground to ground projectile range will be maximum at $\theta = 45^\circ$



Kaam ka Dabba ($a_x = 0$, ground to ground)

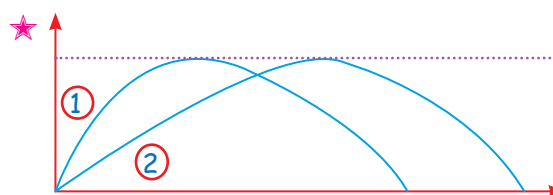
$$T = \frac{2u \sin \theta}{g} = \frac{2U_y}{a_y}$$

$$H_{\max} = \frac{(u \sin \theta)^2}{2g} = \frac{U_y^2}{2a_y}$$

$$R = \frac{u^2 \sin 2\theta}{g} = U_x T = \frac{2U_x U_y}{a_y}$$

★ If vertical velocity same $\Rightarrow T, h_{\max}$, same.

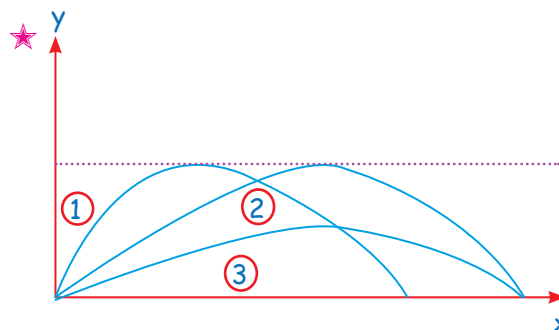
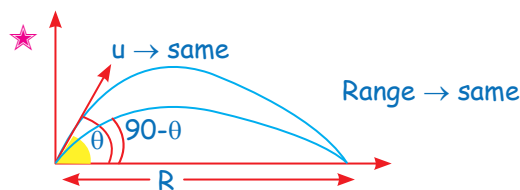
★ If $U \rightarrow$ fix, at θ and $90-\theta \Rightarrow$ Range Same



$H_{\max} \rightarrow$ same

$U_y \rightarrow$ same

$$u_1 \sin \theta = u_2 \sin \theta_2$$



$$(H_{\max})_1 = (H_{\max})_2 > (H_{\max})_3$$

$$T_1 = T_2 > T_3$$

$$(U_y)_1 = (U_y)_2 > (U_y)_3$$

$$R_1 < R_2 = R_3$$

Q. A particle is projected from ground with speed at an angle θ with horizontal such that its maximum height is half of range. Find θ

Sol. $h_{\max} = 1/2 R$

$$\frac{u^2 \sin^2 \theta}{2g} = \frac{1}{2} \frac{u^2 \sin 2\theta}{g}$$

$$\sin^2 \theta = \sin 2\theta$$

$$\sin^2 \theta = 2 \sin \theta \cos \theta$$

$$\tan \theta = 2$$

$$\theta = \tan^{-1}(2)$$

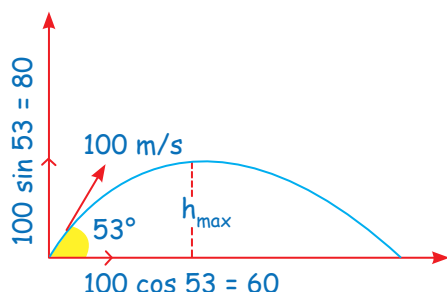
Q. Two particle are projected with same speed at angle 30° & 60° Find Ratio of their max height, range and time of flight respectively.

Sol. (1) $\frac{(h_{\max})_1}{(h_{\max})_2} = \frac{\left(\frac{u^2 \sin^2 30}{2g}\right)}{\left(\frac{u^2 \sin^2 60}{2g}\right)} = \frac{(1/2)^2}{(\sqrt{3}/2)^2} = \frac{1}{3}$

(2) $\frac{R_1}{R_2} = \frac{\left(\frac{u^2 \sin(2 \times 30)}{g}\right)}{\left(\frac{u^2 \sin(2 \times 60)}{g}\right)} = \frac{\sin 60}{\sin 120} = \frac{\sqrt{3}/2}{\sqrt{3}/2} = 1$

(3) $T = \frac{2u \sin \theta}{g} \therefore \frac{T_1}{T_2} = \frac{\sin 30}{\sin 60} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q. A particle is projected with velocity 100m/s at angle of 53° with the horizontal, answer following parts.



1. Find initial velocity, acc

Sol. $\vec{u} = 60\hat{i} + 80\hat{j}$

2. Acceleration.

$\vec{a} = g$ downside

$\vec{a} = 0\hat{i} - 10\hat{j}$

3. Find time of flight

Sol. $T = 8 + 8 = 16$ sec.

4. Range = ?

Sol. $60 \times 16 = 960$

5. Find velocity at

Sol.

$$t = 2 \quad v_y = 60\hat{j} \quad \vec{v} = 60\hat{i} + 60\hat{j}$$

$$t = 8 \quad v_y = 0\hat{j} \quad \vec{v} = 60\hat{i} + 0\hat{j}$$

$$t = 10 \quad v_y = -20\hat{j} \quad \vec{v} = 60\hat{i} - 20\hat{j}$$

$$t = 16 \quad v_y = -80\hat{j} \quad \vec{v} = 60\hat{i} - 80\hat{j}$$

6. Find change in momentum for entire path ($m = 1$ kg) (momentum = mass $\times$ velocity)

Sol. $\Delta \vec{P} = \vec{P}_f - \vec{P}_i$

$$\vec{P}_i = m\vec{v}_i = 1(60\hat{i} + 80\hat{j}) \quad \vec{P}_f = 60\hat{i} - 80\hat{j}$$

$$\Delta \vec{P} = \vec{P}_f - \vec{P}_i = 0\hat{i} - 160\hat{j}$$

$$\Delta \vec{P} = -160\hat{j}$$

7. Find $h_{\max}$

Sol. $v^2 = u^2 + 2as$

$$0^2 = (80)^2 + 2(-10) h_{\max}$$

$$h_{\max} = 320$$

8. Find $\vec{V}$ at any time

Sol. $V = u + at$

$$V_y = 80 - 10t$$

$$\vec{V} = 60\hat{i} + (80 - 10t)\hat{j}$$

9. Find time when $\vec{v}$ become perpendicular to $\vec{a}$

Sol. $\vec{V}_i = 60\hat{i} + (80 - 10t)\hat{j}$

$$\vec{a} = -10\hat{j}$$

$$\vec{V}_i \cdot \vec{a} = -10(80 - 10t) = 0$$

$$t = 8$$

10. Find the time when velocity of particles become perpendicular to initial velocity.

Sol. $\vec{V} = 60\hat{i} + 80\hat{j}$

$$\vec{V}_f = 60\hat{i} + (80 - 10t)\hat{j}$$

$$\vec{V}_i \cdot \vec{V}_f = 0$$

$$60 \times 60 + 80(80 - 10t) = 0$$

$$3600 + 6400 - 800t = 0$$

$$t = \frac{10000}{800} = \frac{100}{8} = 12.5$$

11. Find location of particle at $t = 8$

Sol. $t = 8$, $x = 60 \times 8 = 480$

$$y = 80 \times 8 - \frac{1}{2} \times 10 \times 8^2$$

$$y = 640 - 320 = 320$$

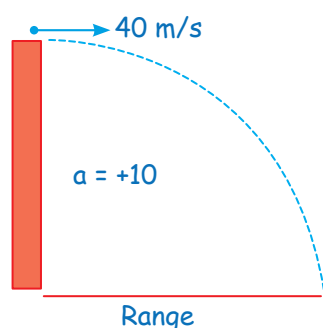
$$(x, y) = (480, 320)$$

Q. A particle is projected horizontally with velocity 40 m/s from top of a tower of height 500 m above ground. Calculate range.

Sol. $u_y = 0$, $a_y = +10$, $s_y = +500$

$$s = ut + \frac{1}{2} at^2$$

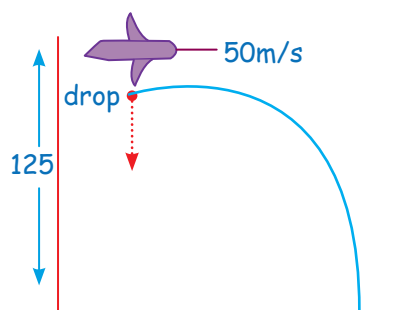
$$500 = 0 + \frac{1}{2} \times 10 \times t^2$$



$$t = 10 \text{ sec.}$$

$$\text{Range} = 40 \times 10 = 400 \text{ m}$$

Q. A body is dropped from plane. Calculate range.

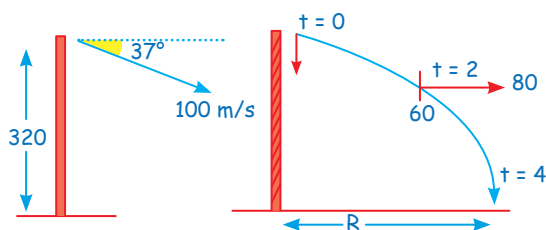


$$\text{Sol. } t = \sqrt{\frac{2h}{a}} = \sqrt{2 \times \frac{125}{10}} = 5$$

$$R = 50 \times 5 = 250 \text{ m}$$

Q. (1) When will particle strike the ground?

(2) Where will particle strike the ground



Sol. (1) $y = \downarrow +ve$ $s = ut + \frac{1}{2} at^2$

$$320 = 60t + \frac{1}{2} 10 at^2$$

$$5t^2 + 60t - 320 = 0$$

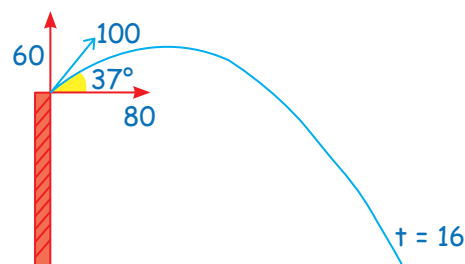
$$\Rightarrow t = 4 \text{ sec}$$

$$(2) \text{ Range} \rightarrow 80 \times 4 = 320$$

Q. A particle is thrown with velocity 100 m/s at an angle of 37° with horizontal from a tower of height 320 m.

(A) Find when particle will hit the ground and where. (Take upward direction is positive)

Sol. (a) $u_y = +60$, $a_y = -10$, $s_y = -320$



$$-320 = 60t - \frac{1}{2} \times 10 \times t^2$$

$$t = 16 \text{ sec}$$

$$R = 80 \times 16$$

(B) Find speed of particle at $t = 16$ sec

$$\vec{v} = 80\hat{i} - 100\hat{j}$$

$$|\vec{v}| = \sqrt{80^2 + 100^2}$$

(C) Find angle made by horizontal at $t = 16$ sec

$$t = 16, \vec{v} = 80\hat{i} - 100\hat{j}$$

$$\tan \alpha = 100/80$$

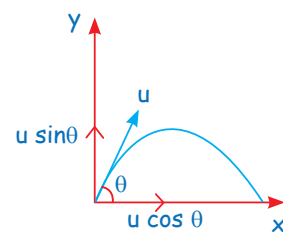
$$\alpha = \tan^{-1} 5/4$$

Equation of Trajectory

$$x = u \cos \theta \cdot t \Rightarrow t = \frac{x}{u \cos \theta}$$

$$y = u \sin \theta t - \frac{1}{2} g t^2$$

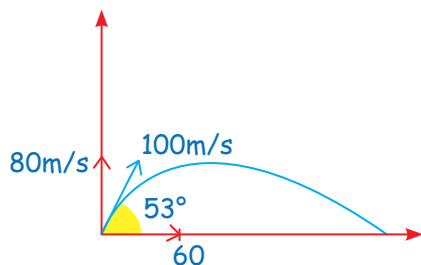
$$y = u \sin \theta \frac{x}{u \cos \theta} - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$



$$y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

$$y = x \tan \theta (1 - x/R)$$

Q. Find equation of trajectory.



Sol. $x = 60t \Rightarrow t = x/60$

$$y = 80t - \frac{1}{2} \times 10 \times t^2$$

$$y = 80 \frac{x}{60} - \frac{1}{2} \times 10 \times \left(\frac{x}{60}\right)^2$$

$$y = \frac{4x}{3} - \frac{x^2}{720}$$

Q. If equation of trajectory is given by

$$y = x\sqrt{3} - \frac{x^2}{20}. \text{ Find } u, \theta.$$

Sol. $y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$ (Formula)

$$\Rightarrow \sqrt{3} = \tan \theta$$

$$\theta = 60^\circ$$

$$\frac{1}{20} = \frac{g}{2u^2 \cos^2 \theta}$$

$$\frac{1}{20} = \frac{10}{2u^2 \cos^2 60}$$

Solve and get $u = 20$

To find the range directly

$$y = x\sqrt{3} - \frac{x^2}{20} = x\sqrt{3} \left(1 - \frac{x}{20\sqrt{3}}\right)$$

Compare this with $y = x \tan \theta (1 - x/R)$

$$\tan \theta = \sqrt{3}, R = 20\sqrt{3}$$

or Put $y = 0$ and get $x = R$

Q. If eqn of trajectory of a projectile from ground to ground is given as $y = 4x - \frac{x^2}{4}$. Find Range = ?

Sol. $y = 4x(1-x/16)$

$$y = x \tan \theta (1-x/R)$$

$$R = 16$$

Q. $y = ax - bx^2$ find θ and Range

Sol. $y = ax(1 - bx^2/ax)$

$$R = a/b$$

$$y = ax(1 - bx/a)$$

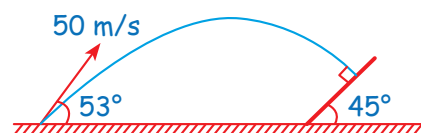
$$\tan \theta = a$$

$$y = 2 \tan \theta (1 - x/R)$$

$$\theta = \tan^{-1} a$$

Q. In following figure particle strike the inclined plane perpendicularly. Find

Time of collision and coordinate of point where particle collide if projection point is origin.

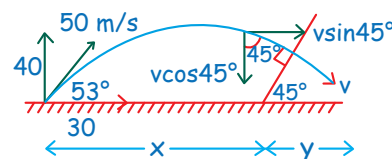


Sol. At the time of collision,

$$v \sin 45^\circ = 50 \cos 53^\circ \Rightarrow v = 30\sqrt{2}$$

$$v_y = 50 \sin 53^\circ - 10T \Rightarrow -30\sqrt{2} \cos 45^\circ = 40 - 10T$$

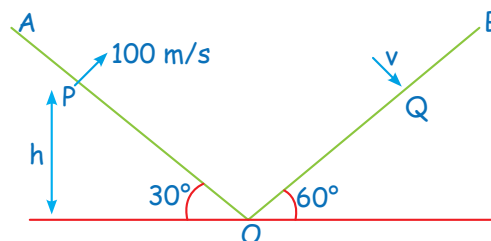
$$\Rightarrow T = 7 \text{ sec}$$



$$x = u \cos \theta \times T = 30 \times 7 = 210$$

$$y = 40 \times 7 - \frac{1}{2} \times 10 \times 7^2 = 35$$

Q. A particle thrown perpendicularly with 100 m/s from an inclined plane and it's strike perpendicularly to another inclined plane as shown in fig. Find with what velocity it will strike the inclined.



Sol. $u \cos 60^\circ = 100 \cos 30^\circ$ (By keeping horizontal velocity)

SKC



Projectile motion ka sabse bada hathiyar horizontal mein velocity constant karna

$$100 \times 1/2 = v \frac{\sqrt{3}}{2} \Rightarrow v = \frac{100}{\sqrt{3}}$$

Q. A particle is moving such that

$$x = 4t^2$$

$$y = 2t^3$$

$$z = 4t^4$$

Find velocity at $t = 2$ sec $\rightarrow$

Sol. $u_x = \frac{dx}{dt} = 8t$

$$\text{Velocity} = 8t\hat{i} + 6t^2\hat{j} + 16t^3\hat{k}$$

$$u_y = \frac{dy}{dt} = 6t^2$$

$$\text{acc} = 8\hat{i} + 12t\hat{j} + 48t^2\hat{k}$$

$$u_z = \frac{dz}{dt} = 16t^3$$

$$\vec{V} = 8t\hat{i} + 6t^2\hat{j} + 16t^3\hat{k}$$

Q. A particle is moving such that $\vec{r} = 3t^2 + 2t^3\hat{j} + 10t\hat{k}$

Find angle between $\vec{v}$ & $\vec{a}$ at $t = 1$ sec.

Sol. $\vec{v} = 6t\hat{i} + 6t^2\hat{j} + 10\hat{k}$

$$\vec{a} = 6\hat{i} + 12t\hat{j} + 0\hat{k}$$

$$\text{At } t = 1, \vec{v} = 6\hat{i} + 6\hat{j} + 10\hat{k}$$

$$\vec{a} = 6\hat{i} + 12\hat{j}$$

$$\vec{a} \cdot \vec{v} = av \cos \theta$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{v}}{av} = \frac{36 + 72 + 0}{\sqrt{6^2 + 12^2} \sqrt{6^2 + 6^2 + 10^2}}$$

RELATIVE MOTION

B $\rightarrow$ 6m/s

A $\rightarrow$ 10 m/s

Velocity of A wrt ground = 10 m/s i

Velocity of B wrt ground = 6 m/s i

$$\text{Velocity of A wrt B} = \vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$$

Velocity of A wrt B = $\vec{v}_{A/B} \equiv$ B ki khopdi par baithkar A ko dekh.

$$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$$

$$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A$$

$$\text{Similarly } \vec{a}_{A/B} = \vec{a}_A - \vec{a}_B$$

★ (A) $\rightarrow$ 6m/s

(B) $\rightarrow$ 8m/s

$$\vec{v}_{B/A} = 8\hat{i} - 6\hat{i} = 2\hat{i}$$

$$\vec{v}_{A/B} = 6\hat{i} - 8\hat{i} = -2\hat{i}$$

★ $\leftarrow$ 10m/s (B)

$\rightarrow$ 6m/s (A)

$$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A = (-10\hat{i} - (6\hat{i})) = -16\hat{i}$$

★ (A) $\rightarrow$ 10m/s

(B) $\rightarrow$ 10m/s

$$\vec{v}_{B/A} = 10 - 10 = 0$$

★ (A) $\rightarrow$ 10m/s

(B) $\uparrow$ 20m/s

$$\vec{v}_{B/A} = 20\hat{j} - 10\hat{i}$$

$$\begin{aligned} \vec{v}_{A/B} &= \vec{v}_A - \vec{v}_B \\ \vec{v}_{B/A} &= \vec{v}_B - \vec{v}_A \\ \vec{v}_{A/B} &= -\vec{v}_{B/A} \end{aligned}$$

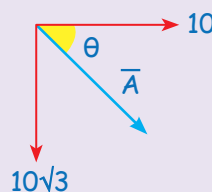
Ab aage humein relative motion, rain man, river boat problems padhne hain, uske liye bahut zaroori hai ki hum vector ko draw karna seekhein. Iske liye, I am attaching a few fundamental things, grab it.

★ $\vec{A} = 10\hat{i} - 10\sqrt{3}\hat{j}$

★ $\vec{v} = -10\hat{i} - 10\sqrt{3}\hat{j}$

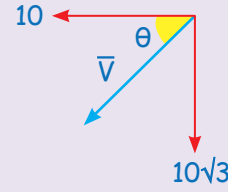
$$\tan \theta = \frac{10\sqrt{3}}{10}$$

$$\theta = 60^\circ$$

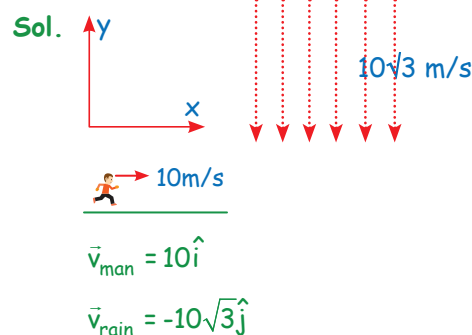


$$\tan \theta = \frac{10\sqrt{3}}{10}$$

$$\theta = 60^\circ$$

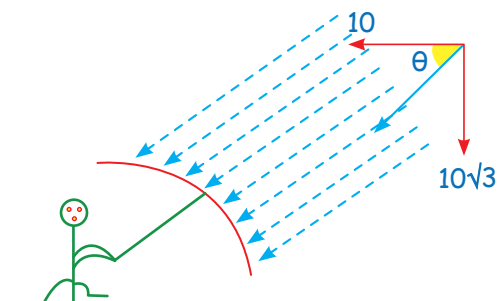


Q. A man is moving along +x-axis (east) with speed 10 m/s & rain is falling vertically downward with speed $10\sqrt{3}$ m/s. In which direction man should hold umbrella to protect himself.



$$\vec{V}_{\text{rain/man}} = \vec{V}_r - \vec{V}_m$$

$$\vec{V}_{r/m} = -10\sqrt{3}\hat{j} - 10\hat{i} = (-10\hat{i} - 10\sqrt{3}\hat{j})$$



Q. A man is moving in east direction with speed 10 m/s in a car. A bird is flying with speed $10\sqrt{3}$ in south direction.

(1) Find velocity of bird observed by man.

Sol. $\vec{V}_{\text{man}} = 10\hat{i}$

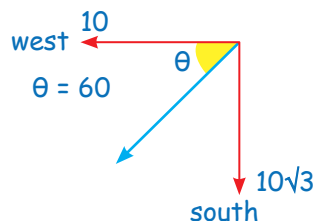
$$\vec{V}_{\text{bird}} = -10\sqrt{3}\hat{j}$$

$$\vec{V}_{b/m} = \vec{V}_b - \vec{V}_m$$

$$= -10\sqrt{3}\hat{j} - 10\hat{i}$$

$$|\vec{V}_{b/m}| = \sqrt{10^2 + (10\sqrt{3})^2}$$

$$= 20 \text{ (} 60^\circ \text{ south of west)}$$

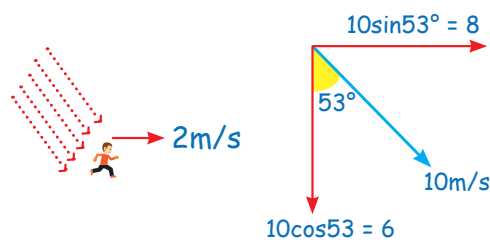


SKC

Rainman prob. jaise ques ke liye ye 4 step follow karo.

1. Sabse pehle velocity of man and rain nikalo.
2. Ab vectorly velocity of rain wrt man $\vec{V}_{r/m}$ nikalo.
3. Ab $\vec{V}_{r/m}$ ko draw karke naya fig. banao.
4. Ab hame pata chal gaya ki man ko rain kahan aati dikh rahi hai so ab chhaata laga do.

Q. Rain is falling with speed 10 m/s at an angle 53° with vertical. A man is moving with speed 2 m/s along east as shown in diagram.



(1) In which direction man should hold umbrella to protect himself

Sol. $\vec{V}_r = 8\hat{i} - 6\hat{j}$

$$\vec{V}_m = 2\hat{i}$$

$$\vec{V}_{r/m} = (8\hat{i} - 6\hat{j}) - (2\hat{i})$$

$$= 6\hat{i} - 6\hat{j}$$

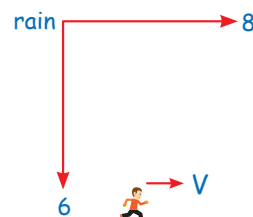
$$\tan \theta = \frac{6}{6}$$

$$\tan \theta = 1$$

$$\tan 45 = 1$$

$$\theta = 45^\circ$$

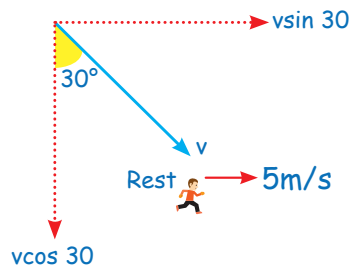
(2) What should be velocity of man so that rain appear falling vertically downward to him?



$$V = 8 \text{ Ans.}$$

Q. To a stationary man, rain appear to be falling at an angle 30° with the vertical. As he start moving with speed of 5 m/s he feels that rain is falling vertically. Find speed of rain.

Sol.



$$v \sin 30 = 5$$

$$v = 10 \text{ speed of rain}$$

Q. A man holding a flag is running with speed 10 m/s along east. If wind speed is $10\sqrt{3}$ m/s along south then. In which direction flag will flutter?

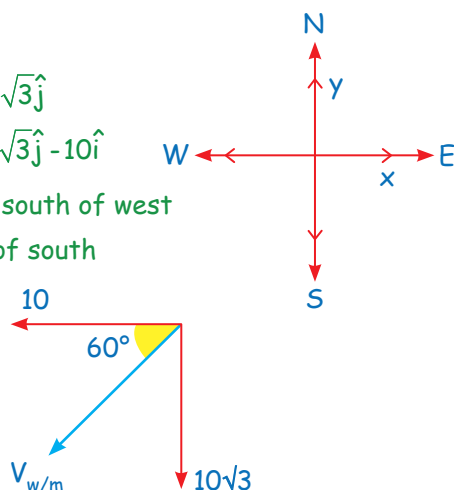
Sol. $\vec{V}_{\text{man}} = 10\hat{i}$

$\vec{V}_{\text{wind}} = -10\sqrt{3}\hat{j}$

$\vec{V}_{w/m} = -10\sqrt{3}\hat{j} - 10\hat{i}$

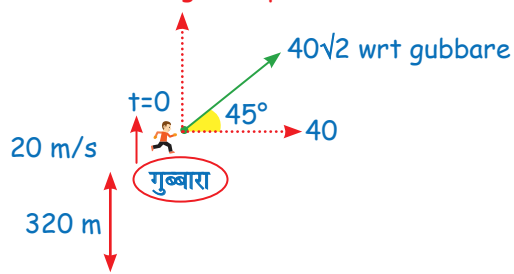
Ans = 60° south of west

30° west of south



Q. A balloon is rising upward with constant velocity 20 m/s. When it is at a height 320 m from ground a particle is projected from balloon with velocity $40\sqrt{2}$ m/s at an angle of 45° with horizontal at $t = 0$.

Find velocity of particle just after projection and time of flight of particle.



$t = 0^+ \quad \vec{V}_p = ? \quad (\text{wrt ground})$

Sol. $\vec{V}_{p/\text{balloon}} = 40\hat{i} + 40\hat{j}$

$\vec{V}_p - \vec{V}_{\text{balloon}} = 40\hat{i} + 40\hat{j}$

$\vec{V}_p - 20\hat{j} = 40\hat{i} + 40\hat{j}$

$\vec{V}_p = 40\hat{i} + 60\hat{j}$

$S = ut + \frac{1}{2}at^2$ (upward +ve)

$-320 = 60t - \frac{1}{2} \times 10 \times t^2$

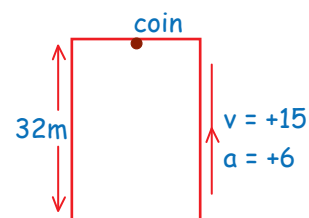
Solve and get $t = 16$ sec

Abe ye to ab 320 m ke tower se patthar fekne wala ques. ban gaya.

SKC

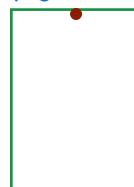
Question mein jab kabhi bhi lift ka, gubbaare ka acc given hota hai, to vo g ko consider karne ke baad given hota hai.

Q. A lift of height 32 m is going up with constant acc $a = 6 \text{ m/s}^2$. When velocity of lift is 15 m/s upward, a coin is drop from ceiling of lift at $t = 0$. Find time when coin will hit the floor.



Sol. Chupchap lift ke andar jakr baith jao wrt lift

$t = 0^+$



(In lift frame or wrt lift)

At $t = 0 \quad \vec{V}_{\text{coin/lift}} = 0$

$\vec{a}_{\text{coin/lift}} = -10 - (+6) = -16$

$\vec{s}_{\text{coin/lift}} = -32$

$S = ut + \frac{1}{2}at^2$

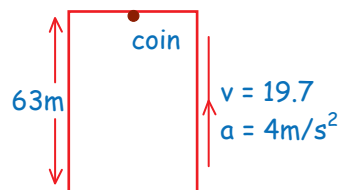
$-32 = 0 - \frac{1}{2} \times 16 \times t^2$

$t = 2 \text{ sec}$

SKC

Aise savalo mein sabse pehle chup chaap lift ke andar jaake baith jao aur coin ki velocity, coin ka acc lift ke respect mein likhkar eqn of motion lagado. Bas ye yaad rakhna agar coin hawa mein hai to ground ke respect mein uska acc neeche g hoga.

Q. At $t = 0$ coin drop. Find time when coin will hit the floor

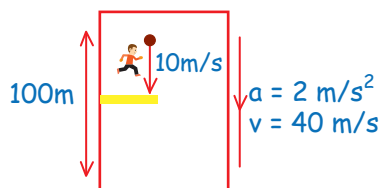


Sol. $\vec{u}_{\text{coin/lift}} = 0$

$\vec{a}_{\text{coin/lift}} = -10 - (+4) = -14$

$$\begin{aligned}\vec{s}_{\text{coin/lift}} &= -63\text{m} \\ s &= ut + \frac{1}{2}at^2 \\ -63 &= 0 + \frac{1}{2} \times (-14)t^2 \\ t &= 3 \text{ sec}\end{aligned}$$

Q. Coin thrown downward at 10m/s w.r.t lift.

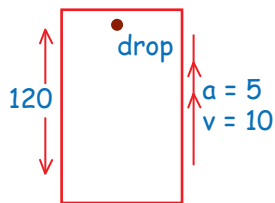


When will coin strike the floor?

Sol. Lift ke andar wrt lift.

$$\begin{aligned}V_{\text{coin/lift}} &= -10 \\ a_{\text{coin/lift}} &= -10 - (-2) = -8 \\ S_{\text{coin/lift}} &= -100 \\ -100 &= -10t + \frac{1}{2}(-8)t^2 \\ 100 &= 10t + 4t^2 \\ 2t^2 + 5t - 50 &= 0 \\ t &= \frac{-5 + \sqrt{25 + 400}}{4} = 4 \text{ sec}\end{aligned}$$

Q. When will coin strike the floor of left.



$$\begin{aligned}\vec{V}_{\text{coin/lift}} &= 0 \\ \vec{a}_{\text{coin/lift}} &= -10 - (+5) = -15 \\ \vec{S}_{\text{coin/lift}} &= -120 \\ -120 &= 0 + \frac{1}{2}(-15)t^2 \\ t &= 4 \text{ sec} \\ \text{lift } t &= 4, S = ?, u = 10, a = 5 \\ s &= 10 \times 4 + \frac{1}{2} \times 5 \times 4^2 = 80\end{aligned}$$

Find displacement of lift w.r.t ground before coin strike the floor?

Displacement of coin in $t = 4$ sec in ground frame

$$\begin{aligned}S &= ut + \frac{1}{2}at^2 \\ s &= 10 \times 4 + \frac{1}{2}(-10) \times 4^2 \\ &= 40 - 80 = -40 \text{ Ans}\end{aligned}$$



Q. Car A and car B start moving simultaneously in the same direction along the line joining them. Car A moves with a constant acceleration $a = 4 \text{ m/s}^2$, while car B moves with a constant velocity $v = 1 \text{ m/s}$. At time $t = 0$, car A is 10 m behind car B. Find the time when car A overtakes car B. How much time A take to overtake.

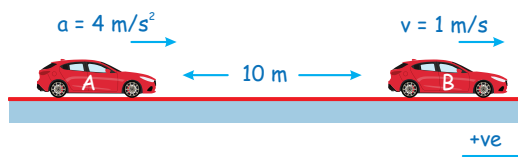
Sol. Given: $u_A = 0$, $u_B = 1 \text{ m/s}$, $a_A = 4 \text{ m/s}^2$ and $a_B = 0$

Assuming car A to be at rest, we have

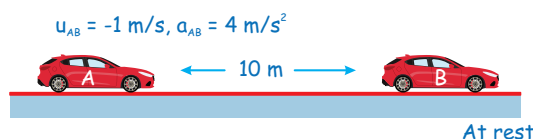
$$u_{AB} = u_A - u_B = 0 - 1 = -1 \text{ m/s}$$

$$a_{AB} = a_A - a_B = 4 - 0 = 4 \text{ m/s}^2$$

Now, the problem can be assumed in simplified form as follow:



Substituting the proper values in equation



$$S = ut + \frac{1}{2}at^2$$

$$\text{we get } 10 = -t + \frac{1}{2}(4)(t^2) \text{ or } 2t^2 - t - 10 = 0$$

Ignoring the negative value, the desired time is 2.5s.

Note: The above problem can also be solved without using the concept of relative motion as under. At the time when A overtakes B,

$$S_A = S_B + 10$$

$$\therefore \frac{1}{2} \times 4 \times t^2 = 1 \times t + 10$$

$$\text{or } 2t^2 - t - 10 = 0$$

Which on solving gives $t = 2.5 \text{ s}$ and -2 s , the same as we found above.

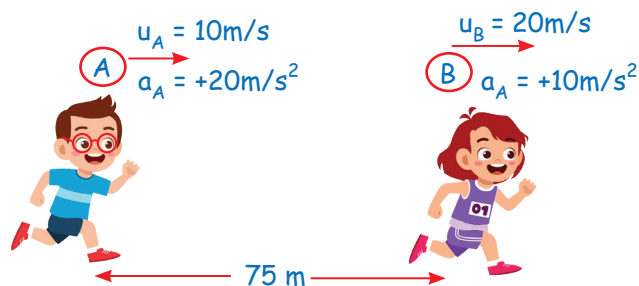
SKC



Catching vaale savalo mein:

- ★ Sabse pehle aage vaale bande par baith jao.
- ★ Baad aage vaale ke respect mein initial v , a , s likho.
- ★ equⁿ of motion thank do.

Q. At $t = 0$ gap between saleemian boy A and girl B is 75 m. Find when boy will catch the girl.



Sol. M-1 (relative wala method, aage wale k upar jakar baith jao)

$$\vec{u}_{A/B} = 10 - 20 = -10$$

$$\vec{a}_{A/B} = 20 - 10 = 10$$

$$s = ut + \frac{1}{2} at^2$$

$$75 = -10t + \frac{1}{2} 10 t^2$$

$$5t^2 - 10t - 75 = 0$$

Solve and get $t = 5$

M-2

$$75 + x_b = x_a$$

$$75 + 20t + \frac{1}{2} \times 10 \times t^2 = 10t + \frac{1}{2} \times 20 \times t^2$$

$$75 + 20t + 5t^2 = 10t + 10t^2$$

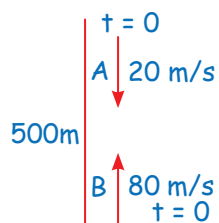
$$5t^2 - 10t - 75 = 0$$

$$t^2 - 2t - 15 = 0$$

$$(t - 5)(t + 3) = 0$$

$$t = 5$$

Q. A & B are thrown as shown simultaneously at $t = 0$ when will they meet?



Sol. $v_{A/B} = -20 - 80 = -100$

$$\vec{a}_{A/B} = -10 - (-10) = 0$$

$$s_{A/B} = -500$$

$$-500 = -100t + \frac{1}{2} \times 0 \times t^2$$

$$t = 5 \text{ sec}$$

Directⁿ

$$\vec{a}_{rel} = 0$$

$$\vec{v}_{rel} = 100$$

$$500 = 100t$$

$$t = 5 \text{ sec}$$

RIVER-BOAT-MAN PROBLEM

Bhai ye JEE-Mains aur NEET ka favourite article hai besharmo ki tarah baar-baar aata rehta hai ise achhe se kar lena.



Q. River is flowing with speed 6 m/s, and $\vec{v}_{man/river}$ is 10 m/s.

Man start swimming from A to B and return back to A. Find time taken



$$\text{Sol. } (A \rightarrow B) t_{A \rightarrow B} = \frac{48}{10+6} = 3 \text{ sec}$$

$$(B \rightarrow A) t_{B \rightarrow A} = \frac{48}{10-6} = 12 \text{ sec}$$

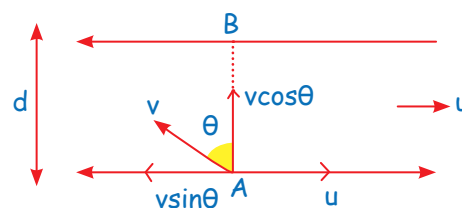
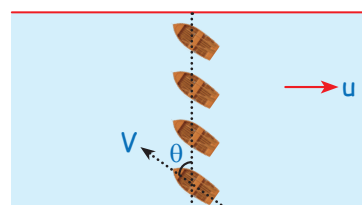
$$\text{total time} = 3 + 12 = 15 \text{ sec}$$

$$\text{Average speed} = \frac{48+48}{15}$$

$$\text{Avg velocity} = 0$$

Case-I

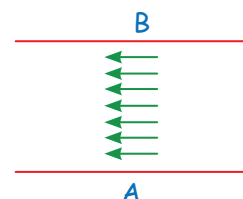
Crossing of river in min distance.



Drift = 0

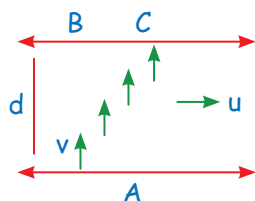
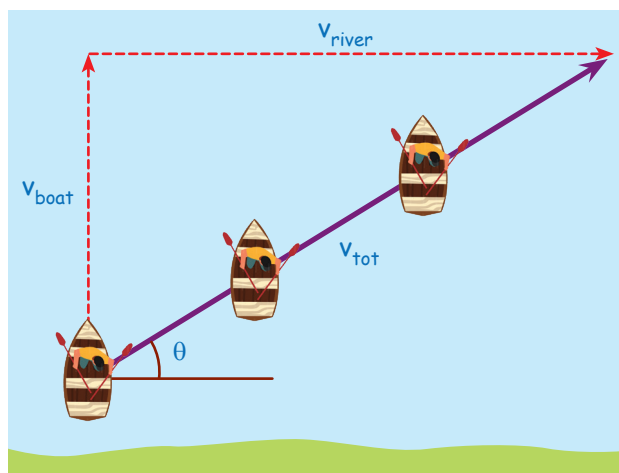
$$V \sin \theta = u$$

$$t = \frac{d}{u \cos \theta}$$



Case-II

Crossing of river in min time

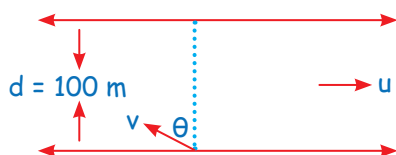


$$\text{Min time} = d/v$$

$$\text{Drift} = CB = ut$$

Q. A river is flowing with velocity 5 m/s wrt ground. A man can swim with velocity 10 m/s wrt river. The width of the river is 100 m.

(1) In which direction man should swim so that he crosses that river in min distance travel also find time taken to cross the river.



Sol. $\vec{v}_{m/r} = v = 10 \text{ m/s}$
 $u = 5 \quad d = 100$

$$v \sin \theta = u$$

$$10 \sin \theta = 5$$

$$\sin \theta = 1/2$$

$$\sin \theta = 30^\circ$$

$$\text{time} = \frac{d}{v \cos \theta} = \frac{100}{10 \cos 30^\circ}$$

(2) If he want to cross the river in min possible time find min time to cross the river & drift

Sol. $t = \frac{100}{10} = 10 \text{ sec}$

$$BC = \text{drift} = 5 \times 10 = 50 \text{ m}$$

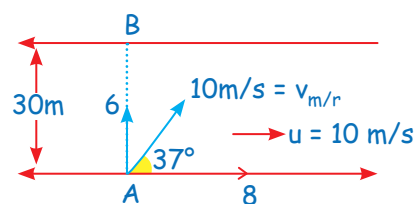
Q. $u = 10\sqrt{3}$, $\vec{v}_{m/r} = 20$

$\theta = ?$ for min distance

Sol. $\sin \theta = \frac{10\sqrt{3}}{20} = \frac{\sqrt{3}}{2}$

$$\theta = 60^\circ$$

Q. In following figure if man is swimming with velocity 10 m/s wrt river at an angle of 37° to the river velocity. Find time taken to cross the river and drift if velocity of river is 10 m/s and width of the river is 30 m.

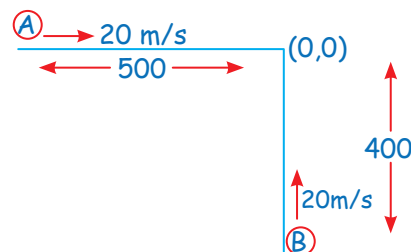


Sol. $t = \frac{30}{6} = 5 \text{ sec}$

Find drift

$$BC = 18 \times 5 = 90 \text{ m} \quad [V_{\text{net}} = 10 + 8 = 18]$$

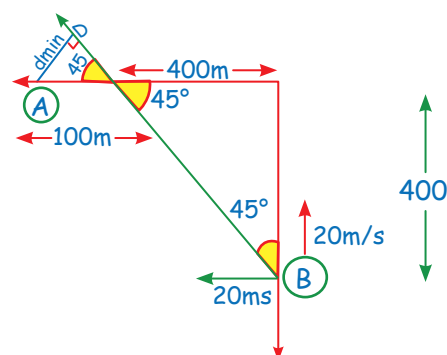
Q. Min distance between the moving particle.



Sol. $\vec{V}_A = -20\hat{i}$

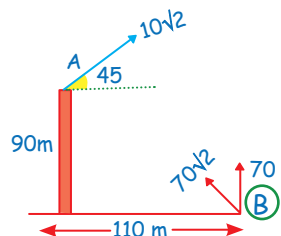
$$\vec{V}_B = 20\hat{j}$$

$$\vec{V}_{B/A} = -20\hat{i} + 20\hat{j}$$

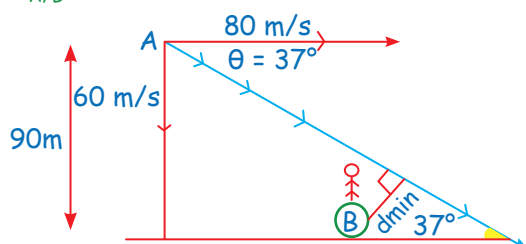


$$AD = d_{\text{min}} = 100 \sin 45^\circ$$

Q. Min Distance between them

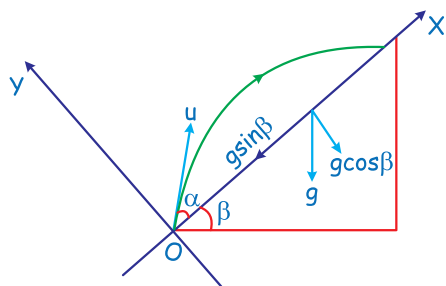


Sol. $\vec{V}_A = 10\hat{i} + 10\hat{j}$
 $\vec{V}_B = -70\hat{i} + 70\hat{j}$
 $\vec{V}_{A/B} = 80\hat{i} - 60\hat{j}$
 $\vec{a}_{A/B} = 0$



PROJECTION ON AN INCLINED PLANE

Case-I: Particle is projected up the incline
 Here α is angle of projection w.r.t. the inclined plane.



$$a_x = -g \sin \beta, \quad u_x = u \cos \alpha$$

$$a_y = -g \cos \beta, \quad u_y = u \sin \alpha$$

Time of flight (T): When the particle strikes the inclined plane, y becomes zero

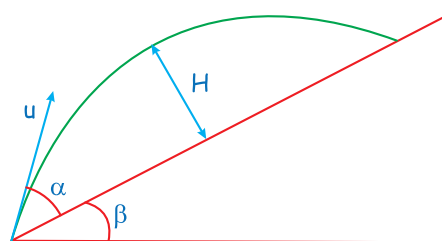
$$y = u_y t + \frac{1}{2} a_y t^2$$

$$\Rightarrow 0 = u \sin \alpha T - \frac{1}{2} g \cos \beta T^2$$

$$\Rightarrow T = \frac{2u \sin \alpha}{g \cos \beta} = \frac{2u_y}{g_y}$$

where $u_{\perp}$ and $g_{\perp}$ are component of u and g perpendicular to the incline.

Maximum Distance from Inclined Plane (H):



When half of the time is elapsed y-coordinate is equal to maximum distance from the inclined plane of the projectile

$$H = u \sin \alpha \left(\frac{u \sin \alpha}{g \cos \beta} \right) - \frac{1}{2} g \cos \beta \left(\frac{u \sin \alpha}{g \cos \beta} \right)^2$$

$$\Rightarrow H = \frac{u^2 \sin^2 \alpha}{2g \cos \beta} = \frac{u_y^2}{2g_y}$$

Range Along the Inclined Plane (R):

When the particle strikes the inclined plane, x coordinate is equal to range of the particle

$$x = u_x t + \frac{1}{2} a_x t^2$$

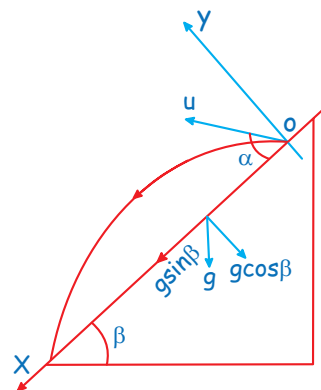
$$\Rightarrow R = u \cos \alpha \left(\frac{2u \sin \alpha}{g \cos \beta} \right) - \frac{1}{2} g \sin \beta \left(\frac{2u \sin \alpha}{g \cos \beta} \right)^2$$

$$\Rightarrow R = \frac{2u^2 \sin \alpha \cos(\alpha + \beta)}{g \cos^2 \beta} \quad (\text{isko ratna mat bs ye})$$

dekho eqn mein value kaise put kr rhe h)

In this case after solving we got range will be maximum when $\alpha = \frac{\pi}{4} - \frac{\beta}{2}$ and Max. Range = $\frac{u^2}{g[1 + \sin \beta]}$

Case-II: Particle is projected down the incline



$$a_x = g \sin \beta \quad u_x = u \cos \alpha$$

$$a_y = -g \cos \beta \quad u_y = u \sin \alpha$$

Time of Flight (T):

When the particle strikes the inclined plane y coordinate becomes zero

$$y = u_y t + \frac{1}{2} a_y t^2 \Rightarrow 0 = u \sin \alpha T - \frac{1}{2} g \cos \beta T^2$$

$$\Rightarrow T = \frac{2u \sin \alpha}{g \cos \beta} = \frac{2u_y}{g_y}$$

Maximum Distance (H):

When half of the time is elapsed y coordinate is equal to maximum distance of the projectile from the plane.

$$H = u \sin \alpha \left(\frac{u \sin \alpha}{g \cos \beta} \right) - \frac{1}{2} g \cos \beta \left(\frac{u \sin \alpha}{g \cos \beta} \right)^2$$

$$H = \frac{u^2 \sin^2 \alpha}{2g \cos \beta} = \frac{u_y^2}{2g_y}$$

Range Along the Inclined Plane (R):

When the particle strikes the inclined plane x coordinate is equal to range of the particle

$$x = u_x t + \frac{1}{2} a_x t^2$$

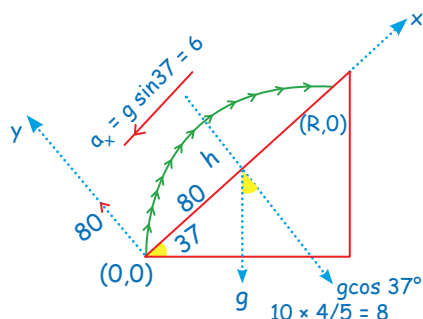
$$\Rightarrow R = u \cos \alpha \left(\frac{2u \sin \alpha}{g \cos \beta} \right) + \frac{1}{2} g \sin \beta \left(\frac{2u \sin \alpha}{g \cos \beta} \right)^2$$

$$R = \frac{2u^2 \sin \alpha \cos(\alpha - \beta)}{g \cos^2 \beta}$$

In this case after solving we got range will be maximum

when $\alpha = \frac{\pi}{4} + \frac{\beta}{2}$ and Max. Range = $\frac{u^2}{g[1 - \sin \beta]}$

Q. A particle is projected with velocity $80\sqrt{2}$ at an angle 45° with inclined plane. If inclined plane makes angle 37° with horizontal. Time of flight and range of particle.



Time of flight = $T = 10 + 10 = 20$ sec

$$\text{Sol. } T = \frac{2u_y}{a_y} = \frac{2 \times 80}{8} = 20 \text{ sec}$$

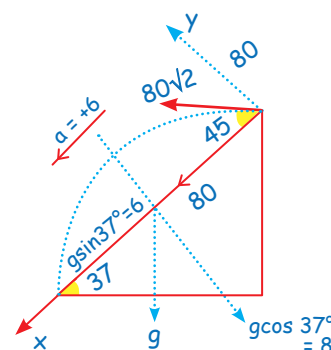
$$R = ut + \frac{1}{2} at^2 \quad (\times \text{ mai})$$

$$R = 80 \times 20 - \frac{1}{2} \times 6 \times (20)^2$$

$$= 1600 - (3 \times 400)$$

$$1600 - 1200 = 400 \text{ m}$$

Q. Repeat the above ques. for following fig.



$$\text{Sol. } T = \frac{2 \times 80}{8} = 20$$

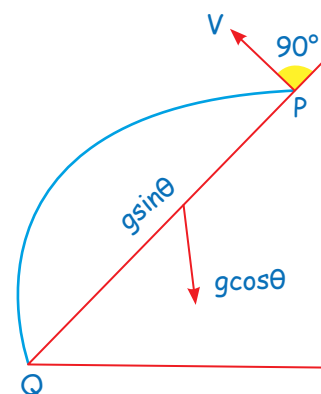
$$\text{Range} = 80 \times 20 + \frac{1}{2} \times 6 \times 20^2$$

$$= 1600 + 3 \times 400$$

$$= 1600 + 1200$$

$$= 2800$$

Q. Time of flight is T find PQ

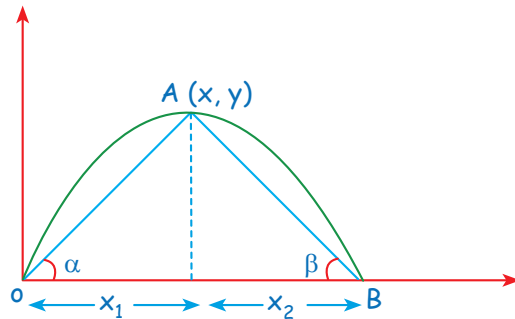


$$\text{Sol. } PQ = \frac{1}{2} g \sin \theta T^2$$

$$= \frac{1}{2} g \sin \theta T \times \frac{2V}{g \cos \theta} \quad \left[T = \frac{2V}{g \cos \theta} \right]$$

$$TV \tan \theta$$

Q. A Projectile Grazing Over a Triangle: A particle is projected over a triangular structure from one of its corners of its horizontal base. Grazing over its top vertex, it falls on other corner of base. If α and β are base angles of triangular structure and θ is angle of projection show that $\tan\theta = \tan\alpha + \tan\beta$.



Sol. $y = x \tan\theta - \frac{1}{2}g \frac{x^2}{u^2 \cos^2 \theta}$

$$y = x \tan\theta \left(1 - \frac{x}{R}\right)$$

$$y = x_1 \tan\theta \left(1 - \frac{x_1}{x_1 + x_2}\right)$$

$$y = \frac{x_1 x_2 \tan\theta}{x_1 + x_2}$$

$$\Rightarrow \tan\theta = y \left(\frac{x_1 + x_2}{x_1 x_2} \right) = \frac{y}{x_2} + \frac{y}{x_1}$$

$$= \tan\alpha + \tan\beta$$